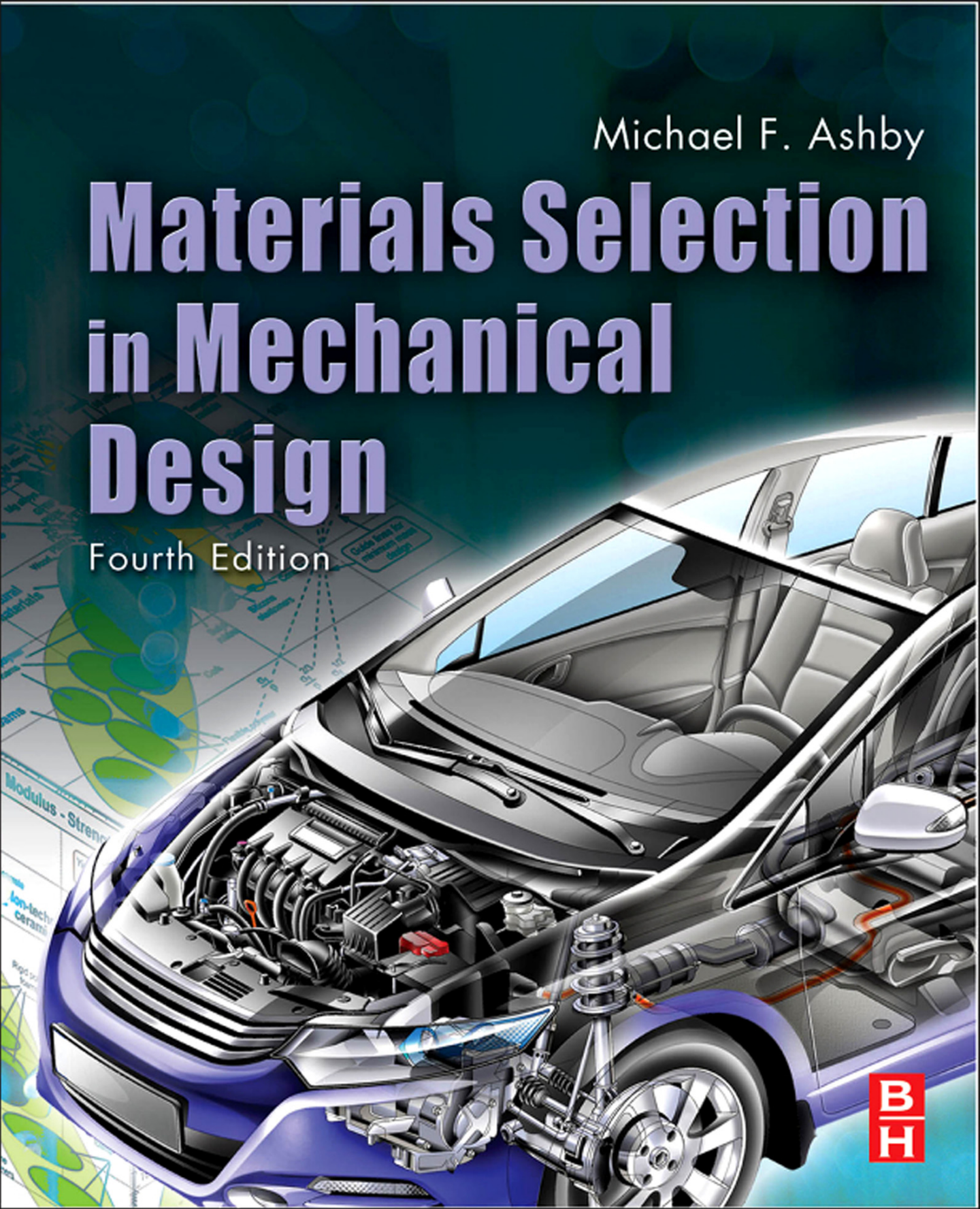
Michael F. Ashby

Materials Selection in Mechanical Design

Fourth Edition



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Fourth Edition

Michael F. Ashby



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Preface

Materials, of themselves, affect us little; it is the way we use them which influences our lives.

Epictetus, AD 50–100, *Discourses*, Book 2, Chapter 5

Materials influenced lives in Epictetus' time and continue to do so today. In his day, the number of materials was small; today it is vast. The opportunities for innovation that materials offer now are equally immense. But advance is possible only if a procedure exists for making a rational choice from the materials on this great menu, and—if they are to be used—a way of identifying ways to shape, join, and finish them. This book develops a systematic procedure for selecting materials and processes, leading to the subset that best matches the requirements of a design. It is unique in the way that the information it contains has been structured. The structure gives rapid access to data and allows the user great freedom in exploring potential choices. The method is implemented in software* to provide greater flexibility.

The approach here emphasizes design with materials rather than materials "science," although the underlying science is used whenever possible to help with the structuring of selection criteria. The first six chapters require little prior knowledge: A first-year grasp of materials and mechanics is enough. The chapters dealing with shape and multiobjective selection are a little more advanced but can be omitted on a first reading. As far as possible, the book integrates materials selection with other aspects of design; the relationships with the stages of design and optimization and with the mechanics of material, are developed throughout. At the teaching level, the book is intended as a text for third- and fourth-year engineering courses on Materials for Design: A 6- to 10-lecture unit can be based on Chapters 1 through 6, 13, and 14; a full 20-lecture course, with project work using the associated software, will require use of the entire book.

* The *CES Edu* materials and process selection platform is a product of Granta Design (*www.grantadesign.com*).

Beyond this, the book is intended as a reference of lasting value. The method, the charts, and the tables of performance indices have application in real problems of materials and process selection; and the table of data and the catalog of "useful solutions" (Appendices A and B) are particularly helpful in modeling—an essential ingredient in optimal design. The reader can use the content (and the software) at increasing levels of sophistication as his or her experience grows, starting with the material indices developed in the book's case studies and graduating to the modeling of new design problems, leading to new material indices and penalty functions, as well as new—and perhaps novel—choices of material. This continuing education aspect is helped by the "Further readings" at the end of each chapter and Appendix E—a set of exercises covering all aspects of the text. Useful reference material is assembled in Appendices A, B, C, and D.

As in any other book, the contents in this one are protected by copyright. Generally, it is an infringement to copy and distribute materials from a copyrighted source. However, the best way to use the charts that are a central feature of the book, for readers to have a clean copy on which they can draw, try out alternative selection criteria, write comments, and so forth; presenting the conclusion for a selected exercise is often most easily done in the same way. Although the book itself is copyrighted, instructors or readers are authorized to make unlimited copies of the charts and to reproduce these for teaching purposes, provided a full reference to their source is given.

ACKNOWLEDGMENTS

Many colleagues have been generous with discussion, criticism, and constructive suggestions. I particularly wish to thank Professor Yves Bréchet of the University of Grenoble in France, Professor Anthony Evans of the University of California at Santa Barbara, Professor John Hutchinson of Harvard University, Professor David Cebon, Professor Norman Fleck, Professor Ken Wallace, Professor John Clarkson, Dr. Hugh Shercliff of the Engineering Department of Cambridge University, Professor Amal Esawi of the American University in Cairo, Professor Ulrike Wegst of Drexel University, Dr. Paul Weaver of the Department of Aeronautical Engineering at the University of Bristol, and Professor Michael Brown of the Cavendish Laboratory in Cambridge, UK.

Mike Ashby

Features of the Fourth Edition

Since publication of the third edition of this book, changes have occurred in the field of materials and their role in engineering, as well as in the way these subjects are taught in university- and college-level courses. There is increasing emphasis on *materials efficiency*—design that uses materials effectively and with as little damage to the environment as possible. All this takes place in a computer-based environment; teaching, too, draws increasingly on computer-based tools. This new edition has been comprehensively revised and reorganized to address these. The presentation has been enhanced and simplified; the figures, many of them new, have been redrawn in full color; worked in-text examples illustrate methods and results in chapters that are not themselves collections of case studies; and additional features and supplements have been added. The key changes are outlined next.

Key changes

- Chapter 1, Introduction, has been completely rewritten and illustrated to develop the history of materials and the evolution of materials in engineering.
- Engineering Design, introduced in Chapter 2, has been edited, with a full revision of all figures.
- Material Properties and Property Charts—a unique feature of the book, which appear in Chapters 3 and 4, have been redrawn in full color.
- Chapter 5 and 6—the central chapters that describe and illustrate selection methods—have been extensively revised with new explanations of the essential selection strategy.
- Chapters 7 and 8 (Multiple Constraints) have been revised, with in-text examples and more illuminating case studies.
- Chapters 9 and 10 (Materials and Shape) have been rewritten for greater clarity, with numerous in-text examples in Chapter 9.

- Chapters 11 and 12, Hybrid Materials, represent a further development of what was in the earlier edition, with a new development of the treatment of sandwich structures and with enhanced case studies.
- Chapters 13 and 14, Processing, contain sections and figures that emphasize the influence of processing on properties.
- Chapter 15, Materials and the Environment, is revised, with improved examples and links to the new information.[1]
- Chapter 16, Industrial Design, is updated and linked to the second edition of the related text[2] on this subject.
- Chapter 17, Forces for Change, has been updated.
- Appendices with Tables of Materials Properties, Useful Solutions, Indices, and Data Sources are updated, enlarged and reillustrated.
- The final appendix contains Exercises that are listed by chapter number.

Material selection charts

Full color versions of a number of the Material Selection Charts presented in this book are available. Samples can be found at *www.grantadesign.com/ashbycharts.htm*. This web page also provides a link to a page where users of CES EduPack (details follow) can download further charts and other teaching resources, including PowerPoint lectures. Although the author retains the copyright for the charts, users of this book are authorized to download, print, and make unlimited copies of those available on the site; in addition, they can be reproduced for teaching purposes (but not for publication), with proper reference to their source.

Instructor's manual and Image Bank

The book ends with a comprehensive set of exercises in Appendix E. Worked-out solutions to the exercises are available, free of charge, to teachers, lecturers, and professors who adopt the book.

The Image Bank provides tutors and lecturers who have adopted this book with PDF versions of the figures contained in it; they can be used for lecture slides and class presentations.

To access the instructor's manual and Image Bank, please visit *www.textbooks.elsevier.com* and follow the onscreen instructions.

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[1] *Materials and the Environment—Eco-informed materials choice* (2009) by M.F. Ashby, Butterworth-Heinemann, ISBN 978-1-85617-608-8.

[2] *Materials and Design—The art and science of materials selection in Product Design*, 2nd edition (2009), by M.F. Ashby and K. Johnson, Butterworth-Heinemann, ISBN 978-1-85617-497-8.

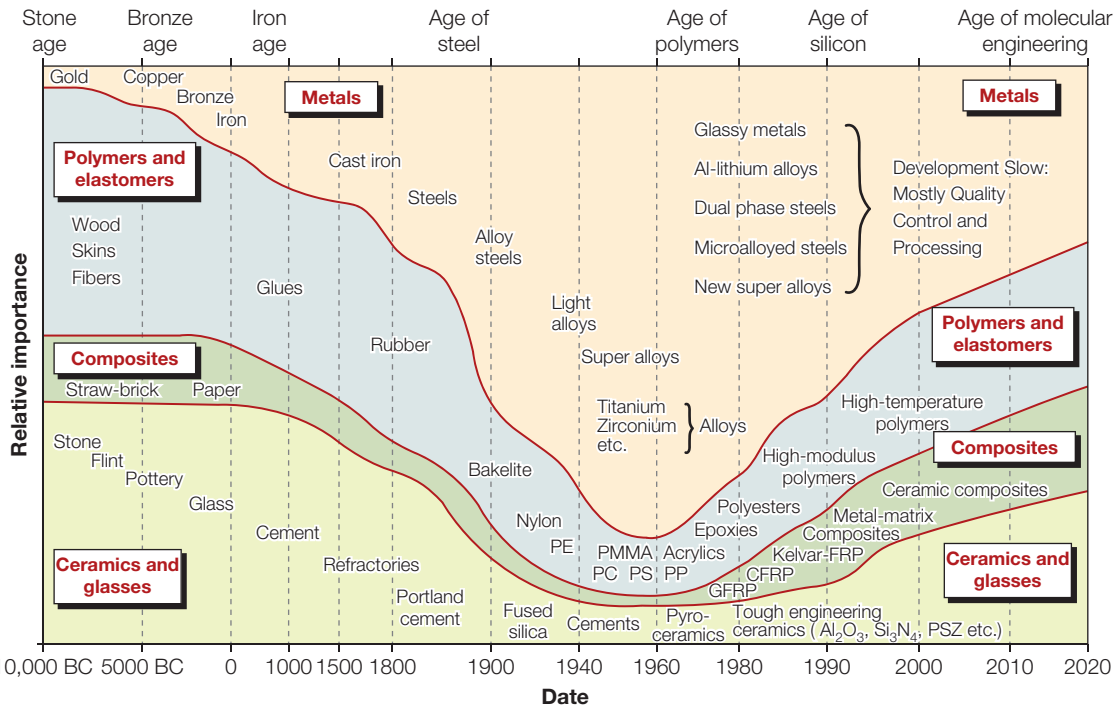
The CES EduPack

The CES EduPack is a widely used software package that implements the methods developed here. The book does not rely on the software, but the learning experience is enhanced by using the two together to create an exciting teaching environment that stimulates exploration, self-teaching, and design innovation. For further information, see the last page of this book or visit *http://www.grantadesign.com/education/*.

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CHAPTER 1

Introduction



The evolution of engineering materials with time. "Relative importance" is based on information contained in the books listed under "Further reading"; plus, from 1960 onward, data for the teaching hours allocated to each material family at U.K. and U.S. universities. The projections to 2020 rely on estimates of material usage in automobiles and aircraft by manufacturers. The time scale is nonlinear. The rate of change is far faster today than at any previous time in history.

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1.1 INTRODUCTION AND SYNOPSIS

"Design" is one of those words that mean all things to all people. Every manufactured thing, from the most lyrical of ladies' hats to the greasiest of gearboxes, qualifies, in some sense or other, as a design. It can mean yet more. Nature, to some, is divine design; to others it is design by natural selection. The reader will agree that it is necessary to narrow the field, at least a little.

This book is about mechanical design and the role of materials in it. Mechanical components have mass; they carry loads; they conduct heat and electricity; they are exposed to wear and to corrosive environments; they are made of one or more materials; they have shape; and they must be manufactured. The book describes how these activities are related.

Materials have had limited design since man first made clothes, built shelters, and waged wars. They still do. But materials and processes to shape them are developing faster now than at any time in history; the challenges and opportunities they present are greater than ever before. This book develops a strategy for confronting such challenges and seizing those opportunities.

1.2 MATERIALS IN DESIGN

Design is the process of translating a new idea or a market need into the detailed information from which a product can be manufactured. Each of its stages requires decisions about the materials of which the product is to be made and the process for making it. Normally, the choice of material is dictated by the design. But sometimes it is the other way around: The new product, or the evolution of the existing one, was suggested or made possible by a new material.

The number of materials available to engineers is vast: 160,000 or more. Although standardization strives to reduce the number, the continuing appearance of new materials with novel and exploitable properties expands the options further. How, then, do engineers choose, from this vast menu, the material best suited to their purpose? Do they rely on their experience? In the past that was how it was done, passing on this precious commodity to apprentices who, much later in their lives, might themselves assume the role of in-house materials guru.

There is no question of the value of experience. But many things have changed in the world of engineering, and all of them work against the success of this model. There is the drawn-out time scale of apprentice-based learning. There is job mobility, meaning that the guru who is here today is usually gone tomorrow. And there is the rapid evolution of materials information, as already mentioned. A strategy that relies on experience is not in tune with today's computer-based environment. We need a *systematic* procedure—one with steps that can be taught quickly, that is robust in the decisions it reaches, that allows computer implementation, and that is compatible with the other established tools of engineering design.

The choice of material cannot be made independently of the choice of process by which the material is to be shaped, joined, and finished. Cost enters the equation, both in the choice of material and in the way the material is processed. So, too, does the influence of material usage on the environment in which we live. And it must be recognized that good engineering design alone is not enough to sell products. In almost everything from home appliances to automobiles and aircraft, the form, texture, feel, color, beauty, and meaning of the product—the satisfaction it gives the person who owns or uses it—are important. This aspect, known confusingly as *industrial design*, is one that, if neglected, can lose markets. Good design works; excellent design also gives pleasure.

Design problems are almost always open-ended. They do not have a unique or "correct" solution, though some solutions will clearly be better than others. They differ from the analytical problems used in teaching mechanics, or structures, or thermodynamics, which generally do have single, correct answers. So the first tool a designer needs is an open mind: a willingness to consider all possibilities. But a net cast widely draws in many different fish. A procedure is necessary for selecting the excellent from the merely good.

This book deals with the materials aspects of the design process. It develops a methodology that, properly applied, gives guidance through the forest of complex choices the designer faces. The ideas of *material* and *process attributes* are introduced. They are mapped on material and process *selection charts* that show the lay of the land, so to speak, and that simplify the initial

survey for potential candidate materials. Real life always involves *conflicting objectives*—minimizing mass while at the same time minimizing cost is an example—requiring the use of *trade-off methods*. The interaction between *material* and *shape* can be built into the method. Taken together, these suggest schemes for expanding the boundaries of material performance by creating *hybrids*—combinations of two or more materials, shapes, and configurations with unique property profiles. None of this can be implemented without *data* for material properties and process attributes: Ways to find them are described. The role of *aesthetics* in engineering design is discussed. The *forces driving change* in the materials world are surveyed, the most obvious of which is the force dealing with environmental concerns. The Appendices contain additional useful information.

The methods lend themselves readily to implementation as computer-based tools; one, the *CES Edu* materials selection platform,[1] is used for many of the case studies and figures in this book. They offer, too, potential for interfacing with tools for computer-aided design, finite element analysis, optimization routines, and product data management software.

All this is found in the following chapters, with case studies illustrating applications. But first, a little history.

1.3 THE EVOLUTION OF ENGINEERING MATERIALS

Throughout history, materials have had limited design. The ages of man are named for the materials he used: stone, bronze, iron. And when a man died, the materials he treasured were buried with him: Tutankhamun in his enameled sarcophagus, Agamemnon with his bronze sword and mask of gold, Viking chieftains in their burial ships—each treasure representing the high technology of their day.

If these men had lived and died today, what would they have taken with them? Their titanium watch, perhaps; their carbon-fiber–reinforced tennis racquet; their metal-matrix composite mountain bike; their shape-memory alloy eyeglass frame with lenses coated with diamond-like carbon; their polyether-ethyl-ketone crash helmet; their carbon nanotube reinforced iPod? This is not the age of one material; it is the age of an immense range of materials. There has never been an era in which their evolution was faster and the range of their properties more varied. The menu of materials has expanded so rapidly that designers who left college 20 years ago can be forgiven for not knowing that many of them exist. But not to know is, for the

[1] Granta Design Ltd., Cambridge, U.K.—*www.grantadesign.com*.

designer, risking disaster. Innovative design often means the imaginative exploitation of the properties offered by new or improved materials. And for the man on the street, the schoolboy even, not to know is to miss one of the great developments of our age: the age of advanced materials.

This evolution and its increasing pace are illustrated on the cover page and, in more detail, in Figure 1.1. The materials of prehistory (before 10,000 BC, the Stone Age) were ceramics and glasses, natural polymers, and composites. Weapons—always the peak of technology—were made of wood and flint; buildings and bridges of stone and wood. Naturally occurring gold and silver were available locally and, through their rarity, assumed great influence as currency, but their role in technology was small. The development of rudimentary thermo-chemistry allowed the extraction of, first, copper and bronze, then iron (the Bronze Age, 4000–1000 BC and the Iron Age, 1000 BC–1620 AD), stimulating enormous advances in technology.[2] Cast iron technology (1620s) established the dominance of metals in engineering; since then the evolution of steels (1850 onward), light alloys (1940s), and special alloys has consolidated their position. By the 1950s, "engineering materials" meant "metals." Engineers were given courses in metallurgy; other materials were barely mentioned.

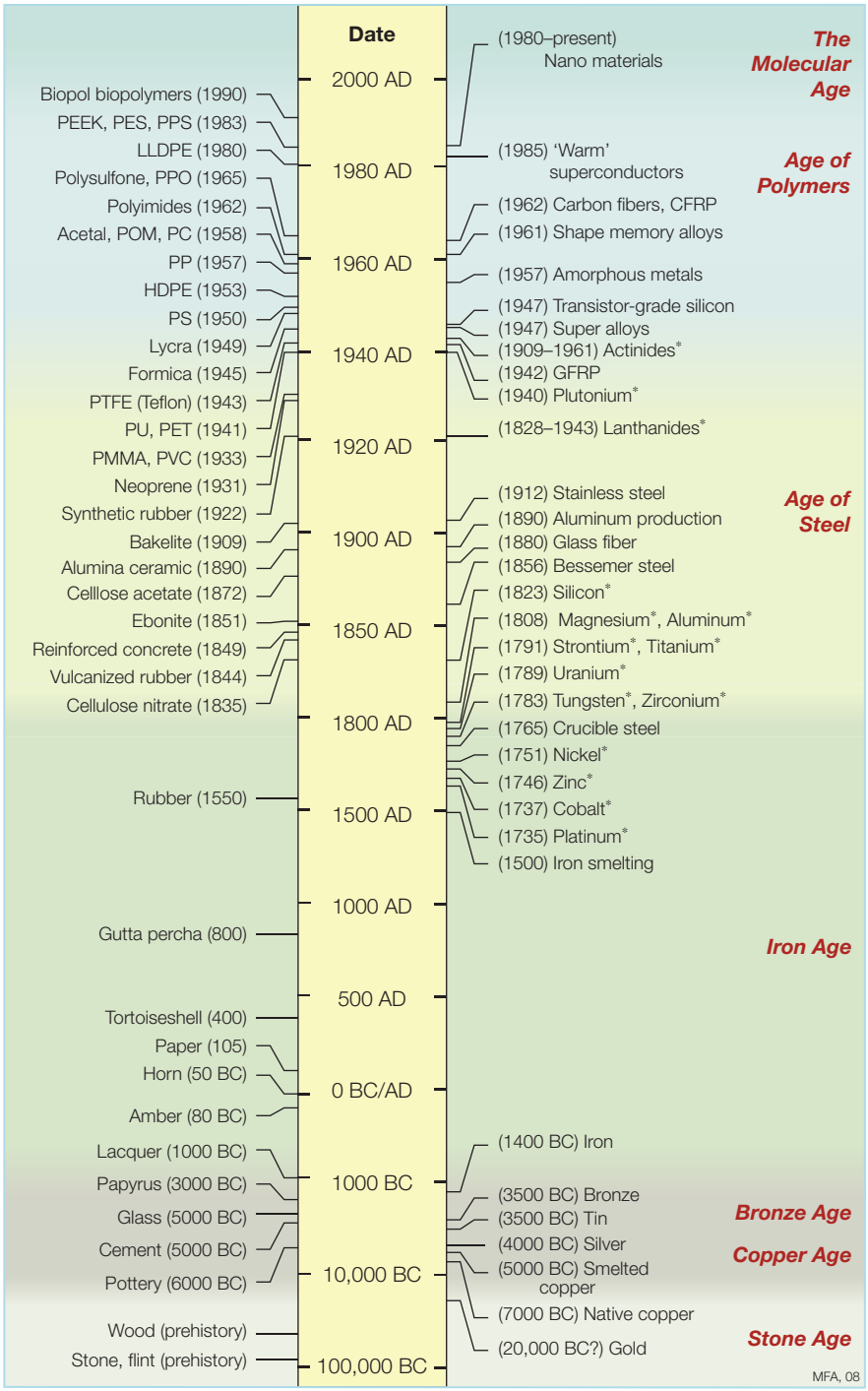
There had, of course, been developments in the other classes of material. Improved cements, refractories, and glasses; and rubber, Bakelite, and polyethylene among polymers; but their share of the total materials market was small. Since 1950 all that has changed. The rate of development of new metallic alloys is now slow; demand for steel and cast iron has in some countries actually fallen.[3] The polymer and composite industries, on the other hand, are growing rapidly, and projections of the growth of production of new high-performance ceramics suggests continued expansion here also.

The material developments documented in the timeline of Figure 1.1 were driven by the desire for ever greater performance. One way of displaying this progression is by following the way in which properties have evolved on *material-property charts*. Figure 1.2 shows one of them—a *strength-density chart*. The oval bubbles plot the range of strength and density of materials; the larger colored envelopes enclose families. The chart is plotted for six successive points in historical time, ending with the present day. The materials of pre-history, shown in (a), cover only a tiny fraction of this strength-density

[2] There is a cartoon on my office door, put there by a student, showing an aggrieved Celt confronting a swordsmith with the words, "You sold me this bronze sword last week and now I'm supposed to upgrade to iron!"

[3] Do not, however, imagine that the days of steel are over. Steel production accounts for 90% of all world metal output, and its unique combination of strength, ductility, toughness, and low price makes it irreplaceable.

FIGURE 1.1
A materials timeline. The scale is nonlinear, with big steps at the bottom, small ones at the top. An asterisk (*) indicates the date at which an element was first identified. Labels without asterisks note the time at which the material became of practical importance.



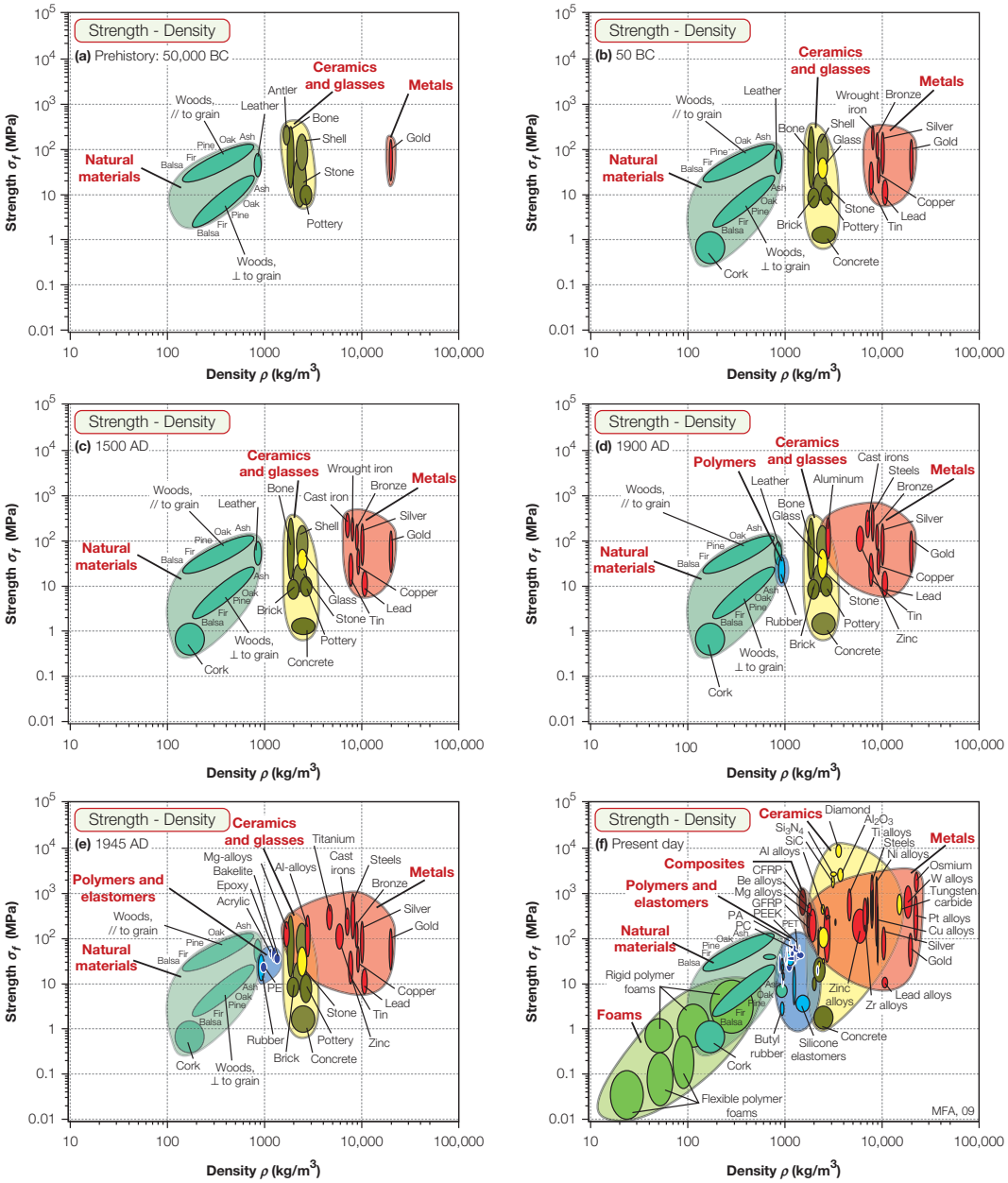


FIGURE 1.2

The progressive filling of material-property space over time (the charts list the date at the top left) showing the way the materials have been developed over time to meet demands on strength and density. Similar time plots show the progressive filling for all material properties.

space. By the time of the peak of the Roman Empire, around 50 BC (b), the area occupied by metals had expanded considerably, giving Rome critical advantages in weaponry and defense. The progress thereafter was slow: 1500 years later (c) not much has changed, although, significantly, cast iron started to appear. Even 500 years after that (d), expansion of the occupied area of the chart is small; aluminum only just starts to creep in. Then things accelerate. By 1945 the metals envelope has expanded considerably and a new envelope—that of synthetic polymers—occupies a significant position. Between then and the present day the expansion has been dramatic. The filled area now starts to approach some fundamental limits (not shown here) beyond which it is difficult to go.

Any slice through material property space (we will encounter many) shows development like this. How can we expand the filled area further? And what would we gain if we did so? These are fascinating questions that will be answered in Chapters 13 and 14. We end this chapter, instead, with a look at the way material developments have been absorbed into products.

1.4 THE EVOLUTION OF MATERIALS IN PRODUCTS

In this section we consider four examples of the changes in the way materials are used, each spanning about 100 years—not much more than a single life time. Bear in mind that in preceding generations, change was far slower. The horse-drawn carriage has a history of 2000 years; the automobile only a little more than 100.

The kettle is the oldest of household appliances and the one found in more homes than any other; there is evidence (not entirely convincing) of a 4000-year-old kettle. Early kettles, heated directly over a fire, were of



FIGURE 1.3
Kettles: cast iron, bronze, polypropylene.

necessity made of materials that could conduct heat well and withstand exposure to an open flame: iron, copper, or bronze (Figure 1.3). Electric kettles, developed in the 1890s, had external heating elements to replace the flame, but were otherwise much like their predecessors. All that changed with the introduction, by the Swan company (1922), of a heating element sealed in a metal tube placed within the water chamber. The kettle body no longer had to conduct heat—indeed for safety and ease of use it was much better made of a thermal and electrical insulator. Today almost all kettles are made of plastic, allowing economic manufacture with great freedom of form and color.

> "Sweeping and dusting are homicidal practices: they consist of taking dust from the floor, mixing it in the atmosphere, and causing it to be inhaled by the inhabitants of the house. In reality it would be preferable to leave the dust alone where it was."

That was a doctor, writing about 100 years ago. More than any previous generation, the Victorians and their contemporaries in other countries worried about dust. They were convinced that it carried disease and that dusting merely dispersed it, when, as the doctor said, it became yet more infectious. Their response was to invent the vacuum cleaner (Figure 1.4).

The vacuum cleaners of that time were human-powered. The materials were, by today's standards, primitive. The cleaner was made almost entirely from natural materials: wood, canvas, leather, and rubber. The only metal parts



FIGURE 1.4

Vacuum cleaners: a hand-powered cleaner from 1880, the Electrolux cylinder cleaner of 1960, and the Dyson centrifugal cleaner of 2010. *Source: Early cleaner (left) courtesy of Worcester News.*



FIGURE 1.5
Cameras: a plate camera from 1900, a Leica from 1960, and a plastic camera from 2006.

were the straps that link the bellows (soft iron) and the can containing the filter (mild steel sheet, rolled to make a cylinder). It reflects the use of materials in that era. Even a car, in 1900, was mostly made of wood, leather, and rubber; only the engine and drive train had to be metal.

The electric vacuum cleaner first appeared around 1908.[4] By 1950 the design had evolved into the cylinder cleaner. Air flow was axial, drawn through the filter by an electric fan. One advance in design was, of course, the electrically driven air pump. But there were others: This cleaner is almost entirely made of metal—the case, the end-caps, the runners, and even the tube to suck up the dust are mild steel. Metals entirely replaced natural materials.

Developments since then have been rapid, driven by the innovative use of new materials. Power and air flow rate are much increased, and dust separation is centrifugal rather than by filtration. This is made possible by higher power density in the motor reflecting improved magnetic materials. The casing is entirely polymeric, and makes extensive use of snap fasteners for rapid assembly. No metal is visible anywhere; even the straight part of the suction tube, which was metal in all earlier models, is now polypropylene.

The optics of a camera are much older than the camera itself (Figure 1.5). Lenses capable of resolving the heavens (Galileo, 1600) or the microscopic (Hooke, 1665) predate the camera by centuries. The key ingredient, of course, was the ability to record the image (Joseph Nicéphore Niépce, 1814). Early cameras were made of wood and constructed with the care and finish of a cabinetmaker. They had well-ground glass lenses and, later, a metal aperture and shutter manufactured by techniques already well developed for watch- and clockmaking.

[4] Inventors: Murray Spengler and William B. Hoover. The second name has become part of the English language, along with the names of such luminaries as John B. Stetson (the hat), S.F.B. Morse (the code), Leo Henrik Baikeland (Bakelite), and Thomas Crapper (the flush toilet).



FIGURE 1.6
Aircraft: the Wright biplane of 1903, the Douglas DC3 of 1935, and the Boeing 787 Dreamliner of 2010.

For the next 90 years photography was practiced by a specialized few. Invention of celluloid-backed film and the cheap box camera around 1900 moved it from a specialized to a mass market, with competition between camera makers to capture a share. Wood, canvas, and leather (with brass and steel only where essential) were quickly replaced by precision-engineered steel bodies and control mechanisms; then from the 1960s on, by aluminum, magnesium, or titanium for low weight, durability, and prestige. Digital technology fractionated the market further. High-end cameras now have optical systems with compound lenses combining glasses with tailored refractive indices, manufactured with the precision and electronic sophistication of scientific instruments. At the other end of the range, point-and-shoot cameras with molded polypropylene or ABS bodies, and acrylic or polycarbonate lenses fill a need.

Perhaps the most dramatic example of the way material usage has changed is found in airframes. Early planes were made of low-density woods (spruce, balsa, and ply), steel wire,[5] and silk. Wood remained the principal structural material of airframes well into the twentieth century, but as planes got larger it became less and less practical. The aluminum airframe, exemplified by the Douglas DC3, was the answer. It provided the high bending stiffness and strength at low weight necessary for scale-up and extended range. Aluminum remained the dominant structural material of civil airliners for the remainder of the twentieth century. By the end of the century, the pressure for greater fuel economy and lower carbon emissions had reached a level that made composites an increasingly attractive choice, despite their higher cost and greater technical challenge. The future of airframes is exemplified by Boeing's 787 Dreamliner (80% carbon-fiber–reinforced plastic by volume), claimed to be 30% lighter per seat than competing aircraft. (See Figure 1.6.)

All this has happened within one lifetime. Competitive design requires the innovative use of new materials and the clever exploitation of their special properties, both engineering and aesthetic. Many manufacturers of kettles, cleaners, and cameras failed to innovate and exploit; now they are extinct. That somber thought prepares us for the chapters that follow, in which we consider what they forgot: the optimum use of materials in design.

[5] "Piano wire"—drawn high-carbon-steel wire—was developed for harpsichords around 1350.

1.5 SUMMARY AND CONCLUSIONS

What do we learn? There is an acceleration in material development and of the ways in which materials are used in products. One of the drivers for change, certainly, is performance: The displacement of bronze by iron in weapons of the Iron Age, and that of wood by aluminum in airframes of the twentieth century, had their origins in the superior performance of the new materials. But performance is not the only factor.

Economics exerts powerful pressures for change: the use of polymers in the jug kettle, the vacuum cleaner, and the cheap camera derives in part from the ease with which polymers can be molded to complex shapes. Technical change in other fields—digital imaging technology, for example—can force change in the way materials are chosen. And there are many more drivers for change that we will encounter in later chapters, among them a concern for the environment, restrictive legislation and directives, aesthetics, and taste.

Engineering materials are evolving faster, and the choice is wider, than ever before. Examples of products in which a new material has captured a market are as common as, well, plastic bottles. Or aluminum cans. Or polycarbonate eyeglass lenses. Or carbon-fiber golf club shafts. It is important in the early stage of design, or of redesign, to examine the full materials menu, not rejecting options merely because they are unfamiliar. That is what this book is about.

1.6 FURTHER READING

The history and evolution of materials

Singer, C. et al. (1954–2001). *A history of technology* (21 volumes). Oxford University Press, ISSN 0307-5451.
A compilation of essays on aspects of technology, including materials.

Delmonte, J. (1985). *Origins of materials and processes*. Technomic Publishing Company, ISBN 87762-420-8.
A compendium of information on when materials were first used and by whom.

Dowson, D. (1998). *History of tribology*. Professional Engineering Publishing Ltd., ISBN 1-86058-070-X.
A monumental work detailing the history of devices limited by friction and wear, and the development of an understanding of these phenomena.

Emsley, J. (1998). *Molecules at an exhibition*. Oxford University Press, ISBN 0-19-286206-5.
Popular science writing at its best: intelligible, accurate, simple, and clear. The book is exceptional for its range. The message is that molecules, often meaning materials, influence our health, our lives, the things we make, and the things we use.

Michaelis, R.R. (Ed.). (1992). Gold: Art, science and technology, and focus on gold. *Interdisciplinary Science Reviews* 17(3, 4), ISSN 0308-0188.
A comprehensive survey of the history, mystique, associations, and uses of gold.

The Encyclopaedia Britannica, 11th Edition (1910). The Encyclopaedia Britannica Company.
Connoisseurs will tell you that in its 11th edition the Encyclopaedia Britannica reached a peak of excellence which has not since been equaled, though subsequent editions are still usable.

Tylecoate, R.F. (1992). *A history of metallurgy* (2nd ed.). The Institute of Materials, ISBN 0-904357-066.
A total-immersion course in the history of the extraction and use of metals from 6000 BC to 1976, told by an author with forensic talent and a love of detail.

Vacuum cleaners

Forty, A. (1986). *Objects of desire—Design in society since 1750*. Thames and Hudson, ISBN 0-500-27412-6. p. 174 et seq.
A refreshing survey of the design history of printed fabrics, domestic products, office equipment and transport systems. The book is mercifully free of eulogies about designers, and focuses on what industrial design does, rather than who did it. The black and white illustrations are disappointing, mostly drawn from the late nineteenth or early twentieth centuries, with few examples of contemporary design.

Cameras

Rosenblum, N. (1989). *A world history of photography* (2nd ed.). Abbeville Press, The University of Michigan, ISBN 1-9-781-558-59054-0.
A study that spans the history of the recorded image, from the camera Lucida to the latest computer technology, including discussion of the aesthetic, documentary, commercial, and technical aspects of its use.

Aircraft

Grant, R. G. (2007). *Flight, the complete history*. Dorling Kindersley Ltd., ISBN 978-1-4053-1768-9.
A lavishly illustrated history, particularly good on the materials of early aircraft.

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CHAPTER 2

The Design Process



Cork removers. (Image courtesy of A-Best Fixture Co., Akron, Ohio.)

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2.1 INTRODUCTION AND SYNOPSIS

We are primarily concerned here with *mechanical design*: the physical principles, the proper functioning, and the production of mechanical systems. This does not mean that we ignore *industrial design*—pattern, color, texture, and (above all) consumer appeal—but that comes later. The optimum starting point in product development is good mechanical design, and the ways in which the selection of materials and processes contribute to it.

Our aim is to develop a methodology for selecting materials and processes that is *design-led*; that is, it uses, as inputs, the functional requirements of the design. To do so we must first look briefly at the design process itself. Like most technical fields, mechanical design is encrusted with its own special jargon, some of it bordering on the incomprehensible. We need very little, but it cannot all be avoided. This chapter introduces some of the words and phrases—the vocabulary—of design, the stages in its implementation, and the ways in which materials selection links with these.

2.2 THE DESIGN PROCESS

The starting point of a design is a *market need* or a *new idea*; the end point is the full *specification* of a product that fills the need or embodies the idea. A need must be identified before it can be met. It is essential to define the need precisely—that is, to formulate a *need statement*, often in this form: "A device is required to perform task X," expressed as a set of *design requirements*. Writers on design emphasize that the need statement should be solution-neutral (that is, it should not imply how the task will be performed) to avoid narrow thinking constrained by preconceptions. Between the need statement and the product specification lie the stages shown in Figure 2.1: *concept*, *embodiment*, and *detailed design*, explained in a moment.

The product itself is called a *technical system*. A technical system consists of *subassemblies* and *components*, put together in a way that performs the required task, as in the breakdown of Figure 2.2. It is like describing a cat (the system) as made up of one head, one body, one tail, four legs, and so on (the subassemblies), each composed of components: femurs, quadriceps, claws, fur. This decomposition is a useful way to analyze an existing design, but it is not of much help in the design process itself, that is, in devising new designs. Better, for this purpose, is one based on the ideas of systems analysis, which considers the inputs, flows, and outputs of information, energy, and materials, as in Figure 2.3.

This design converts the inputs into the outputs. An electric motor, for example, converts electrical into mechanical energy; a forging press takes

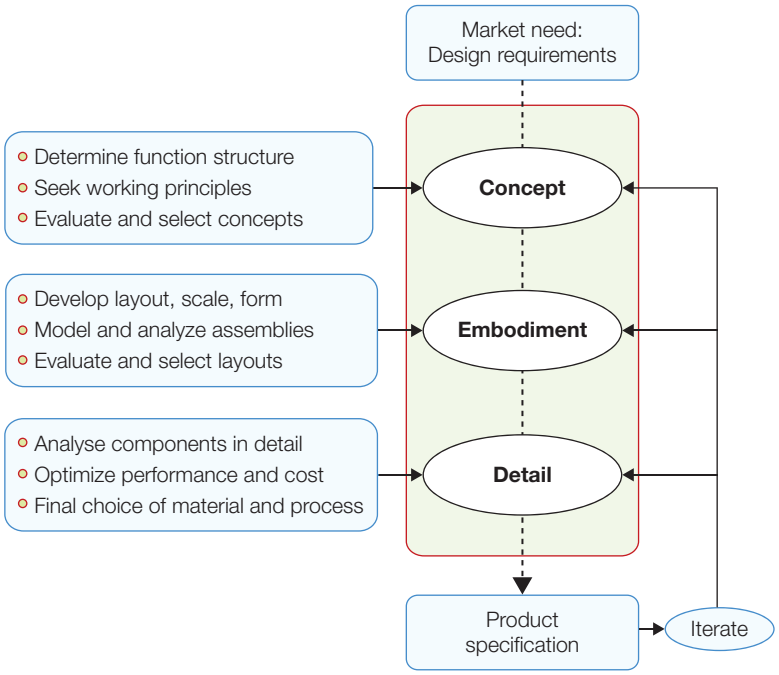


FIGURE 2.1
The design flow chart. The design proceeds from the identification of a *market need*, clarified as a set of *design requirements*, through *concept*, *embodiment*, and *detailed analysis* to a *product specification.*

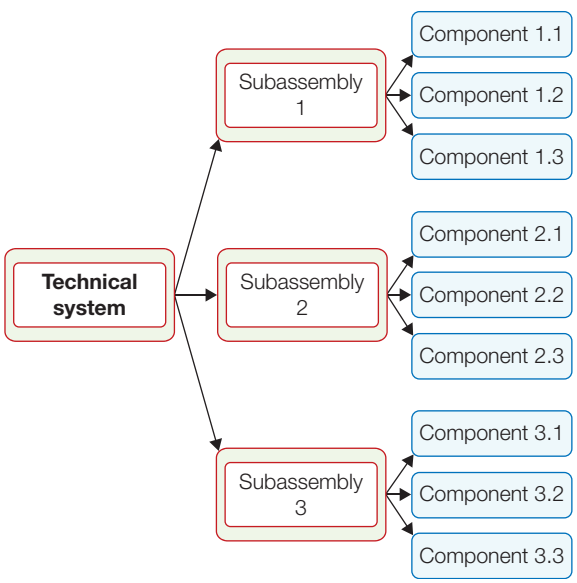


FIGURE 2.2
The analysis of a technical system as a breakdown into assemblies and components. Material and process selection is at the component level.

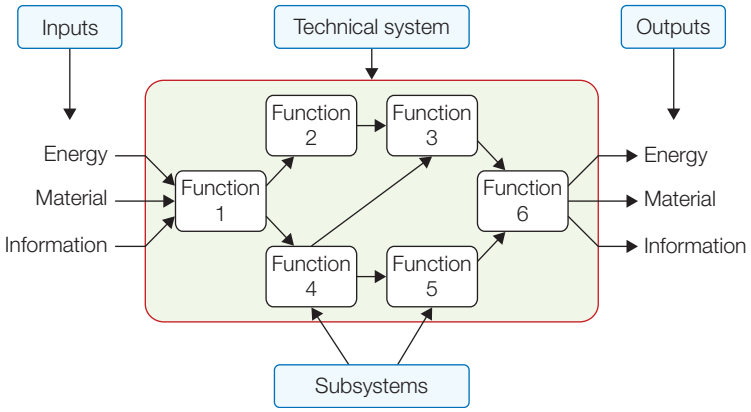


FIGURE 2.3

The function structure is a systems approach to the analysis of a technical system, seen as transformation of energy, materials, and information (signals). This approach, when elaborated, helps structure thinking about alternative designs.

and reshapes material; a burglar alarm collects information and converts it to noise. In this approach, the system is broken down into connected subsystems, each of which performs a specific function, as shown in Figure 2.3. The resulting arrangement is called the *function structure* or *function decomposition* of the system. It is like describing a cat as an appropriate linkage of a respiratory system, a cardio-vascular system, a nervous system, a digestive system, and so on. Alternative designs link the unit functions in alternative ways, combine functions, or split them. The function structure gives a systematic way of assessing design options.

The design proceeds by developing concepts to perform the functions in the function structure, each based on a *working principle*. At this, the conceptual design stage, all options are open: The designer considers alternative concepts and the ways in which these might be separated or combined. The next stage, embodiment, takes the promising concepts and seeks to analyze their operation at an approximate level. This involves sizing the components and selecting materials that will perform properly in the ranges of stress, temperature, and environment suggested by the design requirements, examining the implications for performance and cost. The embodiment stage ends with a feasible layout, which is then passed to the detailed design stage. Here specifications for each component are drawn up. Critical components may be subjected to precise mechanical or thermal analysis. Optimization methods are applied to components and groups of components to maximize performance. A final choice of geometry and material is made and the methods of production are analyzed and costed. The stage ends with a detailed production specification.

All that sounds well and good. If only it were so simple. The linear process suggested by Figure 2.1 obscures the strong coupling between the three stages. The consequences of choices made at the concept or the embodiment stages may not become apparent until the detail is examined. Iteration, looping back to explore alternatives, is an essential part of the design process.

Think of each of the many possible choices that *could* be made as an array of blobs in design space, as shown in Figure 2.4. Here C1, C2… are possible concepts, and E1, E2… and D1, D2… are possible embodiments and detailed elaborations of them. The design process becomes one of creating paths and linking compatible blobs until a connection is made from the top (market need) to the bottom (product specification). Some trial paths have dead ends, some loop back. It is like finding a track across difficult terrain—it may be necessary to go back many times to go forward in the end. Once a path is found, it is always possible to make it look linear and logical (and many books do this), but the reality is more like Figure 2.4 than Figure 2.1.

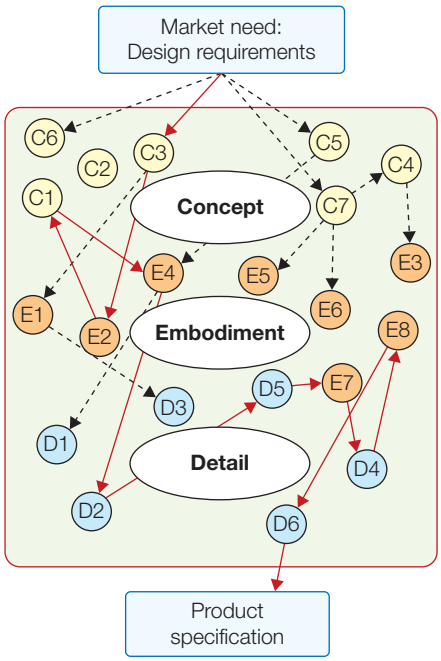


FIGURE 2.4

The convoluted path of design. Here the C-blobs represent concepts; the E-blobs, embodiments of the Cs; and the D-blobs, detailed realizations of the Es. The process is complete when a compatible path from "need" to "specification" can be identified. It is a devious path (the full red line) with back loops and dead ends (the broken lines). This creates the need for tools that allow fluid access to materials information at differing levels of breadth and detail.

Thus a key part of design, and of selecting materials for it, is *flexibility*, the ability to explore alternatives quickly, keeping the big picture as well as the details in mind. Our focus in later chapters is on the selection of materials and processes, where exactly the same need arises. This requires some kind of mapping of the "universes" of materials and processes to allow quick surveys of alternatives while still providing detail when it is needed. The selection charts of Chapter 4 and the methods of Chapter 5 help do this.

Described in the abstract, these ideas are not easy to grasp. An example will help—it comes in Section 2.6. First, a look at types of design.

2.3 TYPES OF DESIGN

It is not always necessary to start, as it were, from scratch. *Original design* does: it involves a new idea or working principle (the ballpoint pen, the compact disc). New materials can offer new, unique combinations of properties that enable original design. Thus high-purity silicon enabled the transistor; high-purity glass, the optical fiber; high coercive-force magnets, the miniature earphone; solid-state lasers the compact disc. Sometimes the new material suggests the new product. Sometimes, instead, the new product demands the development of a new material: Nuclear technology drove the development of a series of new zirconium alloys and low-carbon stainless steels; space technology stimulated the development of light weight composites; gas turbine technology today drives development of high-temperature alloys and ceramic coatings. Original design sounds exciting, and it is. But most design is not like that.

Almost all design is *adaptive* or *developmental.* The starting point is an existing product or product range. The motive for redesigning it may be to enhance performance, to reduce cost, or to adapt it to changing market conditions. Adaptive design takes an existing concept and seeks an incremental advance in performance through a refinement of the working principle. It, too, is often made possible by developments in materials: polymers replacing metals in household appliances; carbon fiber replacing wood in sports equipment. The appliance and the sports equipment markets are fast-moving and competitive. These markets have frequently been won (and lost) by the way in which the manufacturer has adapted the product by exploiting new materials.

Finally, *variant design* involves a change of scale or dimension or detailing without a change of function or the method of achieving it: the scaling up of boilers, or of pressure vessels, or of turbines, for instance. Change of scale or circumstances of use may require change of material: Small boats are made of fiberglass, large ships are made of steel; small boilers are made of

copper, large ones of steel; subsonic planes are made of one alloy, supersonic of another—all for good reasons, as detailed in later chapters.

2.4 DESIGN TOOLS AND MATERIALS DATA

To implement the steps of Figure 2.1, use is made of *design tools*. They are shown as inputs, attached to the left of the main backbone of the design methodology in Figure 2.5. The tools enable the modeling and optimization of a design, easing the routine aspects of each phase. Function modelers suggest viable function structures. Configuration optimizers suggest or refine shapes. Geometric and 3D solid modeling packages allow visualization and create files that can be downloaded to numerically controlled prototyping and manufacturing systems. Optimization, DFM, DFA,[1] and cost estimation software allows manufacturing aspects to be refined. Finite element (FE) and computational fluid dynamics (CFD) packages allow precise mechanical and thermal analysis even when the geometry is complex, deformations are

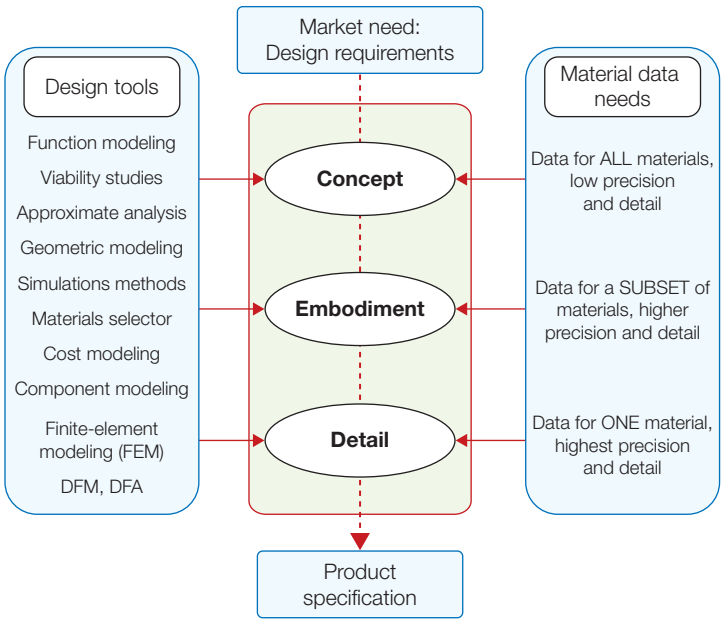


FIGURE 2.5
The design flow chart, showing how design tools and materials selection enter the procedure. Information about materials is needed at each stage, but at very different levels of breadth and precision. Iteration is part of the process.

[1] Design for Manufacture and Design for Assembly.

large, and temperatures fluctuate. There is a natural progression in the use of the tools as the design evolves: approximate analysis and modeling at the conceptual stage; more sophisticated modeling and optimization at the embodiment stage; and precise ("exact"—but nothing is ever that) analysis at the detailed design stage.

Tools for material selection play a major part in each stage of the design. The nature of the data needed in the early stages differs greatly in its level of precision and breadth from that needed later on (Figure 2.5, *right*). At the concept stage, the designer requires approximate property values, but for the widest possible range of materials. All options are open: A polymer may be the best choice for one concept, a metal for another, even though the function is the same. The problem, at this stage, is not precision and detail; it is breadth and speed of access: How can the vast range of data be presented to give the designer the greatest freedom in considering alternatives?

At the embodiment stage the landscape has narrowed. Here we need data for a subset of materials, but at a higher level of precision and detail. These are found in more specialized handbooks and software that deal with a single class or subclass of materials—*metals* or just *aluminum alloys*, for instance. The risk now is that of losing sight of the bigger spread of materials to which we must return if the details don't work out; it is easy to get trapped in a single line of thinking—a single set of "connections" in the sense of Figure 2.4—when other connections may offer a better solution.

The final stage, that of detailed design, requires a still higher level of precision and detail, but for only one or a very few materials. Such information is best found in the datasheets issued by the material producers themselves and in detailed databases for restricted material classes. A given material (polyethylene, for instance) has a range of properties that derive from differences in the ways different producers make it. At the detailed design stage, a supplier must be identified, and the properties of their product used in the design calculations; a product from another supplier may differ. And sometimes even this is not good enough. If the component is a critical one (meaning that its failure could, in some sense or another, be disastrous), then it is prudent to conduct in-house tests to measure the critical properties, using a sample of the material that will be used to make the product itself.

The materials input does not end with the establishment of production. Products fail in service, and failures contain information. It is an imprudent manufacturer who does not collect and analyze data on failures. Often this points to the misuse of a material, one that redesign or reselection can eliminate.

So material choice depends on *function.* But that is not the only constraint.

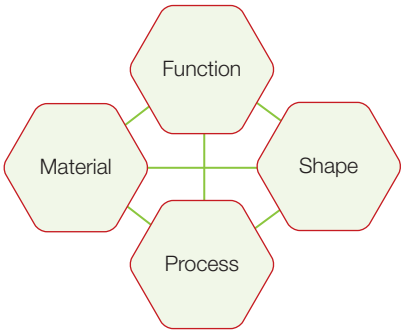


FIGURE 2.6
The central problem of materials selection in mechanical design: the interaction between function, material, process, and shape.

2.5 FUNCTION, MATERIAL, SHAPE, AND PROCESS

The selection of material is tied in with process and shape. To make a shape, the material is subjected to processes that, collectively, are called *manufacture*: these include primary forming processes (e.g., casting and forging), material removal processes (machining, drilling), joining processes (e.g., welding) and finishing processes (e.g., painting or electroplating). Function, material, shape, and process interact (Figure 2.6). Function, as already described, influences material choice. Material choice influences processes through the material's ability to be cast or molded or welded or heat-treated. Process determines shape; size; precision; and, of course, cost. These interactions are two-way: specification of shape restricts the choice of material and process; but equally the specification of process limits the material choice and the accessible shapes. The more sophisticated the design, the tighter the specifications and the greater the interactions.

The interaction between function, material, shape, and process lies at the heart of the material selection process. It is a theme we will return to throughout this book, visiting each of the hexagons of Figure 2.6 in turn. But first we look at a case study to illustrate the design process.

2.6 CASE STUDY: DEVICES TO OPEN CORKED BOTTLES

Wine, like cheese, is one of man's improvements on nature. And as long as humans have cared about wine, people have cared about corks to keep it safely sealed in flasks and bottles. "Corticum… demovebit amphorae…,"

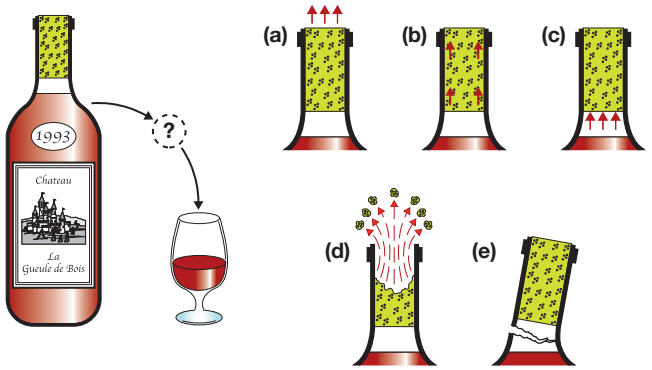


FIGURE 2.7

Left: the market need; a device is sought to allow access to wine contained in a corked bottle. *Right:* five possible concepts, illustrating physical principles, to fill the need.

("Uncork the amphora…)" sang Horace[2] to celebrate the anniversary of his miraculous escape from death by a falling tree. But how did he do it?

A corked bottle creates a market need: the need to gain access to the wine inside. We might state it thus: "A device is required to pull corks from wine bottles." But hold on. The need must be expressed in solution-neutral form, and this is not. The aim is to gain access to the wine; our statement implies that this will be done by removing the cork, and that the cork will be removed by pulling. There could be other ways. So we will try again: "A device is required to allow access to wine in a corked bottle" (Figure 2.7), and one might add, "with convenience, at modest cost, and without contaminating the wine."

Five concepts for doing this are shown on the right of Figure 2.7. In order, the devices act to remove the cork by axial traction (pulling); to remove it by shear tractions; to push it out from below; to pulverize it; and to bypass it altogether by knocking the neck off the bottle.[3]

Numerous devices exist that use the first three of these concepts. The others are used too, though generally only in moments of desperation. We shall eliminate these on the grounds that they might contaminate the wine, and examine the others more closely, exploring working principles. Figure 2.8 shows one example for each of the first three concepts: In the first, a screw

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[2] Horace, Q. 27 BC, *Odes*, Book III, Ode 8, line 10.

[3] A Victorian invention for opening old port, the cork of which may have become brittle with age and alcohol absorption, involved ring-shaped tongs. The tongs were heated red on an open fire, then clamped onto the cold neck of the bottle. The thermal shock removed the neck cleanly and neatly.

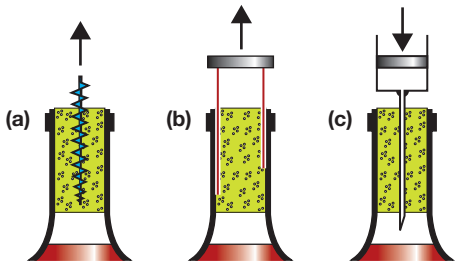


FIGURE 2.8
Working principles for implementing the first three concepts of Figure 2.7. Examples of all of these appear in the cover picture of this chapter.

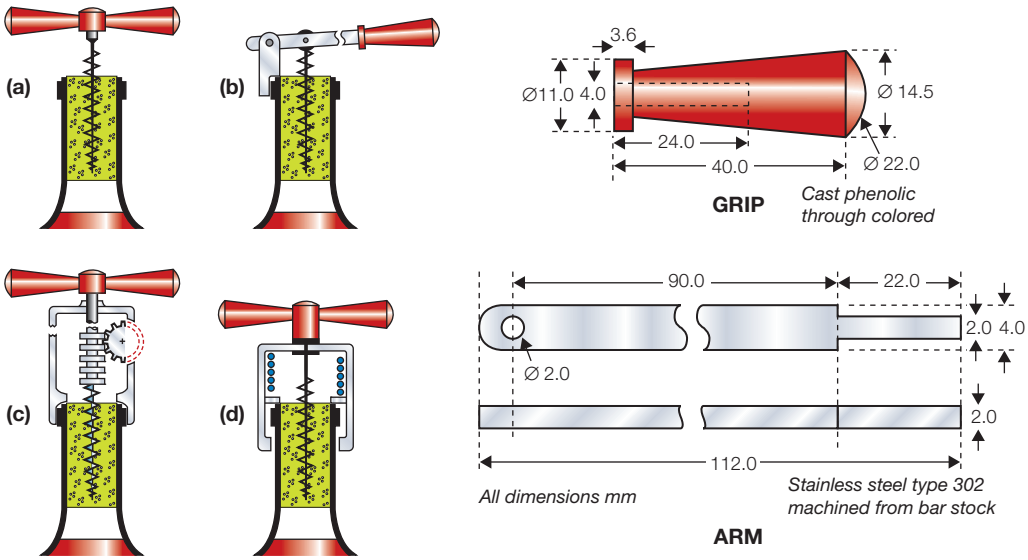


FIGURE 2.9
Left: embodiments: (a) direct pull; (b) lever-assisted pull; (c) gear-assisted pull; (d) spring-assisted pull (a spring in the body is compressed as the screw is driven into the cork). *Right:* detailed design of the lever of embodiment with material choice.

is threaded into the cork to which an axial pull is applied; in the second, slender elastic blades inserted down the sides of the cork apply shear tractions when twisted and pulled; and in the third, the cork is pierced by a hollow needle through which a gas is pumped to push the cork out. Examples of all three appear in the cover picture of this chapter.

Figure 2.9 shows embodiment sketches for devices based on just one concept—that of axial traction. The first is a direct pull; the other three use some sort of mechanical advantage—levered pull, geared pull, and spring-assisted pull. The embodiments identify the *functional requirements*

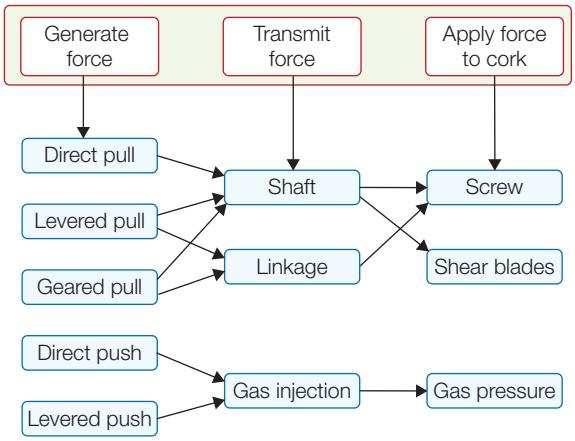


FIGURE 2.10
The function structure and working principles of cork removers.

of each component of the device, which might be expressed in statements such as

- A cheap screw to transmit a prescribed load to the cork
- A light lever (that is, a beam) to carry a prescribed bending moment
- A slender elastic blade that will not buckle when driven between the cork and the bottle neck
- A thin, hollow needle, stiff and strong enough to penetrate a cork

The functional requirements of each component are the inputs to the materials selection process. They lead directly to the *property limits* and *material indices*, as described in Chapter 5, in which we examine procedures with requirements such as "light, strong beam" or "slender, elastic, blade" and use them to identify a subset of materials that perform this function particularly well. The final choice of material and process forms part of the detailed stage of design (Figure 2.9), leading to full specifications to enable manufacture.

We conclude by returning to the idea of function structure. That for the cork remover is sketched in the upper part of Figure 2.10: Generate a force, transmit a force, apply the force to the cork. The alternative designs differ in the working principle by which these functions are achieved, as indicated in the lower part of the figure. Others could be devised by making other links.

2.7 SUMMARY AND CONCLUSIONS

Design is an iterative process. The starting point is a *market need* captured in a set of *design requirements*. *Concepts* for a product that meet the need are devised. If initial estimates and exploration of alternatives suggest that the concept is

viable, the design proceeds to the *embodiment* stage: Working principles are selected, size and layout are decided, and initial estimates of performance and cost are made. If the outcome is successful, the designer proceeds to the *detailed design* stage: optimization of performance, full analysis of critical components, preparation of detailed production drawings (usually as CAD files), specifications of tolerance, precision, assembly, and finishing methods.

Materials selection enters at each stage, but at different levels of breadth and precision. At the conceptual stage all materials and processes are potential candidates, requiring a procedure that allows rapid access to data for a wide range of each, although without the need for great precision. The preliminary selection passes to the embodiment stage, the calculations and optimizations of which require information at a higher level of precision and detail. They eliminate all but a small shortlist of candidate materials and processes for the final, detailed stage of the design. For these few candidates, highest-quality data are required.

Data exist that meet the needs of all these levels. Each level requires its own data management scheme, described in the following chapters. The management system must be design-led, yet must recognize the richness of choice and embrace the complex interaction between the material, its shape, the process by which it is given that shape, and the function it is required to perform. And it must allow rapid iteration—back-looping when a particular path proves to be unprofitable. Tools now exist to help with all of this. We will meet one—the CES materials and process selection platform—later in this book.

But given this complexity, why not opt for the safe bet: Stick to what you used before? Many have chosen that option. Few are still in business.

2.8 FURTHER READING

A chasm exists between books on design methodology and those on materials selection: each largely ignores the other. The book by French is remarkable for its insights, but the word "material" does not appear in its index. Pahl and Beitz have near-biblical standing in the design camp, but their text is heavy going. Ullman and Cross take a more relaxed approach and are easier to digest. The books by Budinski and Budinski, by Charles, Crane, and Furness, and by Farag present the materials case well, but are less good on design. Lewis illustrates material selection through case studies, but does not develop a systematic procedure. The best compromise, perhaps, is Dieter.

General texts on design methodology

Cross, N. (2000). *Engineering design methods* (3rd ed.). Wiley, ISBN 0-471-87250-4.
A durable text describing the design process, with emphasis on developing and evaluating alternative solutions.

Dieter, G.E., & Schmidt, L.C. (2009). *Engineering design* (4th ed.). McGraw-Hill, ISBN 978-0-07-283703-2.
A clear introduction from authors with a strong materials background.

French, M.J. (1985). *Conceptual design for engineers*. The Design Council, London, and Springer, ISBN 0-85072-155-5 and 3-540-15175-3.
The origin of the "Concept–Embodiment–Detail" block diagram of the design process. The book focuses on the concept stage, demonstrating how simple physical principles guide the development of solutions to design problems.

Pahl, G., & Beitz, W. (1997). *Engineering design* (2nd ed.). Translated by K. Wallace & L. Blessing, The Design Council, London, and Springer Verlag, ISBN 0-85072-124-5 and 3-540-13601-0.
The Bible—or perhaps more exactly the Old Testament—of the technical design field, developing formal methods in the rigorous German tradition.

Ullman, D.G. (1992). *The mechanical design process*. McGraw-Hill, ISBN 0-07-065739-4.
An American view of design, developing ways in which an initially ill-defined problem is tackled in a series of steps, much in the way suggested by Figure 2.1 of the present text.

Ulrich, K.T., & Eppinger, S.D. (1995). *Product design and development*. McGraw-Hill, ISBN 0-07-065811-0.
A readable, comprehensible text on product design, as taught at MIT. Many helpful examples but almost no mention of materials.

General texts on materials selection in design

Ashby, M., Shercliff, H., & Cebon, D. (2010). *Materials: Engineering, science, processing and design* (2nd ed.). Butterworth-Heinemann, ISBN 978-1-85617-895-2. North American edition: ISBN 13:978-1-85617-743-6.
An introductory text introducing ideas that are developed more fully in the present text.

Askeland, D.R., & Phulé, P.P. (2006). *The science and engineering of materials* (5th ed.). Thomson, ISBN 0-534-55396-6.
A well-established materials text that deals well with the science of engineering materials.

Budinski, K.G., & Budinski, M.K. (2010). *Engineering materials, properties and selection* (9th ed.). Prentice Hall, ISBN 978-0-13-712842-6.
Like Askeland, this is a well established materials text that deals well with both material properties and processes.

Callister, W.D. (2010). *Materials science and engineering: An introduction* (8th ed.). John Wiley & Sons, ISBN 978-0-470-41997-7.
A well-established text taking a science-led approach to the presentation of materials teaching.

Charles, J.A., Crane, F.A.A., & Furness, J.A.G. (1997). *Selection and use of engineering materials* (3rd ed.). Butterworth-Heinemann, ISBN 0-7506-3277-1.
A materials science rather than a design-led approach to the selection of materials.

Dieter, G.E. (1999). *Engineering design, a materials and processing approach* (3rd ed.). McGraw-Hill, ISBN 9-780-073-66136-0.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

Farag, M.M. (2008). *Materials and process selection for engineering design* (2nd ed.). CRC Press, Taylor and Francis, ISBN 9-781-420-06308-0.
A materials science approach to the selection of materials.

Lewis, G. (1990). *Selection of engineering materials*. Prentice-Hall, ISBN 0-13-802190-2.
A text on material selection for technical design, based largely on case studies.

Shackelford, J.F. (2009). *Introduction to materials science for engineers* (7th ed.). Prentice Hall. ISBN 978-0-13-601260-4.
A well-established materials text with a design slant.

And on corks and corkscrews

The Design Council. (1994). Teaching aids program EDTAP DE9. The Design Council, London.

McKearin, H. (1973). On "stopping," bottling and binning. *International Bottler and Packer*, (April), 47–54.

Perry, E. (1980). *Corkscrews and bottle openers*. Shire Publications.

Watney, B.M., & Babbige, H.D. (1981). *Corkscrews*. Sotheby's Publications, ISBN 0-85667-113-4.

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CHAPTER 3

Engineering Materials and Their Properties

Acrylonitrile butadiene styrene (ABS)

General properties

Density	1100	-	1200	kg/m^3
Price	2.1	-	2.5	USD/kg

Mechanical properties

Young's modulus	1.1	-	2.9	GPa
Yield strength	19	-	51	MPa
Tensile strength	28	-	55	MPa
Elongation	15	-	40	%
Hardness—Vickers	5.6	-	15	HV
Fatigue strength at 10^7 cycles	11	-	22	MPa
Fracture toughness	1.2	-	4.3	$MPa.m^{1/2}$

Thermal properties

Glass temperature	360	-	400	K
Maximum service temperature	340	-	350	K
Minimum service temperature	150	-	200	K
Thermal conductivity	0.19	-	0.34	W/m.K
Specific heat capacity	1400	-	1900	J/kg.K
Thermal expansion coefficient	85	-	230	$10^{-6}/°C$

Electrical properties

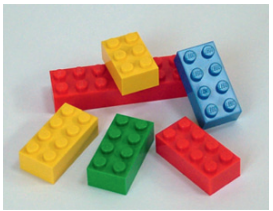
Electrical resistivity	3.3×10^{21}	-	3×10^{22}	μohm.cm
Dielectric constant	2.8	-	3.2	
Dielectric loss tangent	0.003	-	0.007	
Dielectric strength	14	-	22	MV/m

Eco-properties

Embodied energy	91	-	1e2	MJ/kg
CO_2 footprint	3.3	-	3.6	kg/kg

Part of a record of the properties of an engineering material, ABS.

The material



ABS (Acrylonitrile-butadiene-styrene) is tough, resilient, and easily molded. It is usually opaque, although some grades can now be transparent, and it can be given vivid colors. ABS-PVC alloys are tougher than standard ABS and, in self-extinguishing grades, are used for the casings of power tools. The picture shows that ABS allows detailed moldings, accepts color well, and is nontoxic and tough enough to survive the worst that children can do.

Typical uses

Safety helmets; camper tops; automotive instrument panels and other interior components; pipe fittings; home-security devices and housings for small appliances; communications equipment; business machines; plumbing hardware; automobile grilles; wheel covers; mirror housings; refrigerator liners; luggage shells; tote trays; mower shrouds; boat hulls; large components for recreational vehicles; weather seals; glass beading; refrigerator breaker strips; conduit; pipe for drain-waste-vent (DWV) systems.

Tradenames

Claradex, Comalloy, Cycogel, Cycolac, Hanalac, Lastilac, Lupos, Lustran ABS, Magnum, Multibase, Novodur, Polyfabs, Polylac, Porene, Ronfalin, Sinkral, Terluran, Toyolac, Tufrex, Ultrastyr.

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3.1 INTRODUCTION AND SYNOPSIS

Materials, one might say, are the food of design. This chapter presents the menu: the materials shopping list. A successful product—one that performs well, is good value for money, and gives pleasure to the user—uses the best materials for the job, and fully exploits their potential and characteristics. It brings out their flavor, so to speak.

The families of materials—metals, polymers, ceramics, and so forth—are introduced in Section 3.2. What do we need to know about them if we are to design with them? That is the subject of Section 3.3, in which distinctions are drawn between various types of materials information. But it is not, in the end, a *material* that we seek; it is a certain *profile of properties*—the one that best meets the needs of the design. Properties are the currency of the materials world. They are the bargaining chips—the way you trade off one material against another. The properties important in thermo-mechanical design are defined briefly in Section 3.4. It makes boring reading. The reader who is confident in the definitions and units of moduli, strengths, damping capacities, thermal and electrical conductivities, and the like, may wish to skip this, using it for reference, when needed, for the precise meaning and units of the data in the property charts that come later. Don't, however, skip Section 3.2. It sets up the classification structure that is used throughout the book. The chapter ends, in the usual way, with a summary.

3.2 THE FAMILIES OF ENGINEERING MATERIALS

It is conventional to classify the materials of engineering into the six broad families shown in Figure 3.1: metals, polymers, elastomers, ceramics, glasses, and hybrids. The members of a family have certain features in

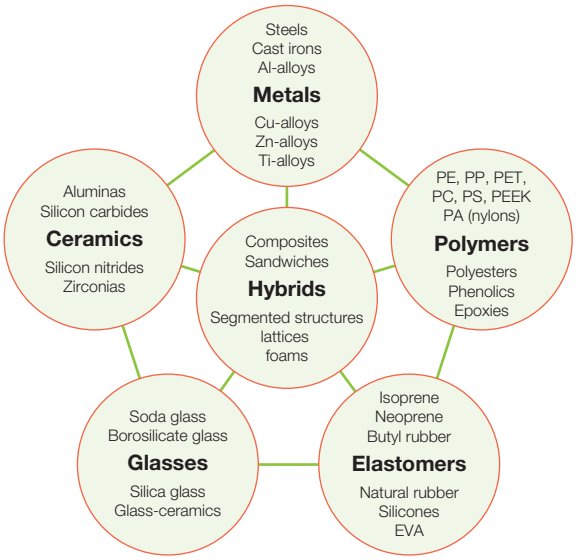


FIGURE 3.1
The menu of engineering materials. The basic families of metals, ceramics, glasses, polymers, and elastomers can be combined in various geometries to create hybrids.

common: similar properties, similar processing routes, and, often, similar applications.

Metals are stiff. They have relatively high elastic moduli. Most, when pure, are soft and easily deformed. They can be made strong by alloying and by mechanical and heat treatment, but they remain ductile, allowing them to be formed by deformation processes. Certain high-strength alloys (spring steel, for instance) have ductilities as low as 1%, but even this is enough to ensure that the material yields before it fractures and that fracture, when it occurs, is of a tough, ductile type. Partly because of their ductility, metals are prey to fatigue and of all the classes of material, they are the least resistant to corrosion.

Ceramics, too, have high moduli, but unlike metal, they are brittle. Their "strength" in tension means the brittle fracture strength; in compression it is the brittle crushing strength, which is about 15 times greater. And because ceramics have no ductility, they have a low tolerance for stress concentrations (like holes or cracks) or for high-contact stresses (at clamping points, for instance). Ductile materials accommodate stress concentrations by deforming in a way that redistributes the load more evenly, and because of this, they can be used under static loads within a small margin of their yield strength. Ceramics cannot. Brittle materials always have a wide scatter in strength, and the strength itself depends on

the volume of material under load and the time over which it is applied. So ceramics are not as easy to design with as metals. Despite this, they have attractive features. They are stiff, hard, and abrasion-resistant (hence their use for bearings and cutting tools); they retain their strength to high temperatures; and they resist corrosion well.

Glasses are noncrystalline ("amorphous") solids. The most common are the soda-lime and borosilicate glasses familiar as bottles and ovenware, but there are many more. Metals, too, can be made noncrystalline by cooling them sufficiently quickly. The lack of crystal structure suppresses plasticity, so, like ceramics, glasses are hard, brittle, and vulnerable to stress concentrations.

Polymers are at the other end of the spectrum. They have moduli that are low, roughly 50 times lower than those of metals, but they can be strong—nearly as strong as metals. A consequence of this is that elastic deflections can be large. They creep, even at room temperature, meaning that a polymer component under load may, with time, acquire a permanent set. And their properties depend on temperature so that a polymer that is tough and flexible at 20°C may be brittle at the 4°C of a household refrigerator, yet may creep rapidly at the 100°C of boiling water. Few have useful strength above 200°C. Some polymers are mainly crystalline, some are amorphous (noncrystalline), some are a mix of crystalline and amorphous—transparency goes with the amorphous structure. If these aspects are allowed for in the design, the advantages of polymers can be exploited. And there are many. When combinations of properties, such as strength per unit weight, are important, polymers can compete with metals. They are easy to shape. Complicated parts performing several functions can be molded from a polymer in a single operation. The large elastic deflections allow the design of polymer components that snap together, making assembly fast and cheap. And by accurately sizing the mold and precoloring the polymer, no finishing operations are needed. Polymers resist corrosion (paints, for instance, are polymers) and have low coefficients of friction. Good design exploits these properties.

Elastomers are long-chain polymers above their glass-transition temperature, T_g. The covalent bonds that link the units of the polymer chain remain intact, but the weaker Van der Waals and hydrogen bonds that, below T_g, bind the chains to each other, have melted. This gives elastomers unique properties: Young's moduli as low as 10^{-3} GPa (10^5 times less than that typical of metals) increase with temperature (all other solids show a decrease), and have enormous elastic extension. Their properties differ so much from those of other solids that special tests have evolved to characterize them. This creates a problem: If we wish to select materials by prescribing a desired attribute profile, as we do later in this book, then a prerequisite is a set of attributes common to all materials.

To overcome this, we use a common set of properties in the early stages of design, estimating approximate values for anomalies like elastomers. Specialized attributes, representative of one family only, are stored separately; they are for use in the later stages.

Hybrids are combinations of two or more materials in a predetermined configuration and scale. They combine the attractive properties of the other families of materials while avoiding some of their drawbacks. Their design is the subject of Chapters 11 and 12. The family of hybrids includes fiber and particulate composites, sandwich structures, lattice structures, foams, cables, and laminates; almost all the materials of nature—wood, bone, skin, and leaf—are hybrids. Fiber-reinforced composites are, of course, the most familiar. Most of those at present available to the engineer have a polymer matrix reinforced by fibers of glass, carbon, or Kevlar (an aramid). They are light, stiff, and strong, and they can be tough. They, and other hybrids using a polymer as one component, cannot be used above 250°C because the polymer softens, but at room temperature their performance can be outstanding. Hybrid components are expensive, and they are relatively difficult to form and join. So, despite their attractive properties, the designer will use them only when the added performance justifies the added cost. Today's growing emphasis on high performance and fuel efficiency provides increasing drivers for their use.

These, then, are the material families. What do we need to know about them?

3.3 MATERIALS INFORMATION FOR DESIGN

The engineer, in selecting materials for a developing design, needs data for the materials' properties. Engineers are often conservative in their choice, reluctant to consider materials with which they are unfamiliar, and with good reason. Data for the old, well-tried materials are established, reliable, and easily found. Data for newer, emerging, materials may be incomplete or untrustworthy. Yet innovation is often made possible by new materials. So it is important to know how to judge data quality.

If you're going to design something, what sort of materials information do you need? Figure 3.2 draws relevant distinctions. On the left a material is tested and the *data* are captured. But these raw data—unqualified numbers—are, for our purposes, useless. To make data useful requires statistical analysis. What is the mean value of the property when measured on a large batch of samples? What is the standard deviation? Given these, it is possible to calculate *allowables*: values of properties that, with a given certainty (say, one part in 10^6) can be guaranteed. Material texts generally present test data; by contrast, data in most engineering handbooks are allowables.

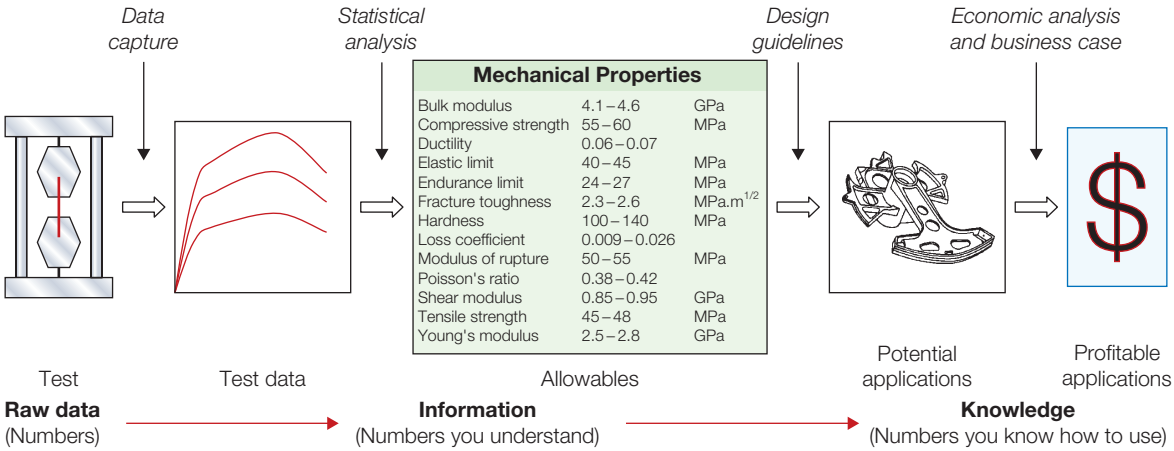


FIGURE 3.2

Types of material information. We are interested here in the types found in the center of this schematic: structured data for design "allowables" and the characteristics of a material that relate to its ability to be formed, joined, and finished; records of experience with its use; and design guidelines for its use.

One can think of data with known precision and provenance as *information*. Information can generally be reported as tables of numbers, as yes/no statements or as rankings: that is, it can be *structured*. Many attributes that can be structured are common to all materials; all have a density, an elastic modulus, a strength, a thermal conductivity. Structured information can be stored in a database and—since all materials have values—it is the starting point for selecting between them. The cover picture of this chapter shows part of a record for the polymer ABS with structured data on the left, reported as ranges that derive from differences in the way different producers make it.

This is a step forward, but it is not enough. To design with a material, you need to know its real character, its strengths, and its weaknesses. How do you shape it? How do you join it? Who has used it before and for what? Did it fail? Why? This information exists in handbooks, is documented as design guidelines, and is reported in failure analyses and case studies. It consists largely of text, graphs and images, and while certain bits of it may be available for one material, for another they may not. It is messier, but it is essential in reaching a final selection. We refer to this supporting information as *documentation*. The image and text on the right of the ABS cover are examples of documentation.

There is more. Material uses are subject to standards and codes. These rarely refer to a single material but to classes or subclasses. For a material to be used in contact with food or drugs, it must carry FDA approval or the equivalent. Metals and composites for use in U.S. military aircraft must have military

specification approval. To qualify for best-practice design for the environment, material usage must confirm to ISO 14040 guidelines. And so forth. This, too, is a form of documentation (Table 3.1). The ensemble of information about a material, structured and unstructured, constitutes *knowledge*.

There is yet more (Figure 3.2, *right*). To succeed in the marketplace, a product must be economically viable and compete successfully, in terms of performance, consumer appeal, and cost, with the competition. All of these

Table 3.1 Basic Design-Limiting Material Properties and Their Usual SI Units*

Class	Property	Symbol and Units
General	Density	ρ (kg/m^3 or Mg/m^3)
	Price	C_m ($/kg)
Mechanical	Elastic moduli (Young's, shear, bulk)	E, G, K (GPa)
	Yield strength	σ_y (MPa)
	Tensile (ultimate) strength	σ_{ts} (MPa)
	Compressive strength	σ_c (MPa)
	Failure strength	σ_f (MPa)
	Hardness	H *(Vickers)*
	Elongation	ε (–)
	Fatigue endurance limit	σ_e (MPa)
	Fracture toughness	K_{1c} (MPa.m$^{1/2}$)
	Toughness	G_{1c} (kJ/m^2)
	Loss coefficient (damping capacity)	η (–)
	Wear rate (Archard) constant	K_AMPa^{-1}
Thermal	Melting point	T_m (°C or K)
	Glass temperature	T_g (°C or K)
	Maximum service temperature	T_{max} (°C or K)
	Minimum service temperature	T_{min} (°C or K)
	Thermal conductivity	λ (W/m.K)
	Specific heat	C_p (J/kg.K)
	Thermal expansion coefficient	α (K^{-1})
	Thermal shock resistance	ΔT_s (°C or K)
Electrical	Electrical resistivity	ρ_e (Ω.m or $\mu\Omega$.cm)
	Dielectric constant	ε_r (–)
	Breakdown potential	V_b (10^6 V/m)
	Power factor	P (–)
Optical	Refractive index	n (–)
Eco-properties	Embodied energy	H_m (MJ/kg)
	Carbon footprint	CO_2 (kg/kg)

** Conversion factors from metric to imperial and cgs units appear inside the back and front covers of this book.*

depend on material choice and the way the material is processed. Much can be said about this, but not here; for now the focus is on structured data and documentation.

That's the essential background. Now for the properties themselves.

3.4 MATERIAL PROPERTIES AND THEIR UNITS

Each material can be thought of as having a set of attributes or properties. The combination that characterizes a given material is its *property profile*. Property profiles are assembled by systematic testing. In this section we scan the nature of the tests and the definition and units of the properties (see Table 3.1). Property values are listed in Appendix A. Units are given here in the SI system. Conversion factors to other systems are printed on the inside front and back cover of the book.

General properties

The *density*, ρ (units: kg/m^3), is the mass per unit volume. We measure it today as Archimedes did: by weighing in air and in a fluid of known density.

The *price*, C_m (units: \$/kg), spans a wide range. Some cost as little as \$0.2/kg, others as much as \$1,000/kg. Prices, of course, fluctuate, and they depend on the quantity you want and on your status as a "preferred customer" with your chosen vendor. Despite this uncertainty, it is useful to have an approximate price in the early stages of material selection.

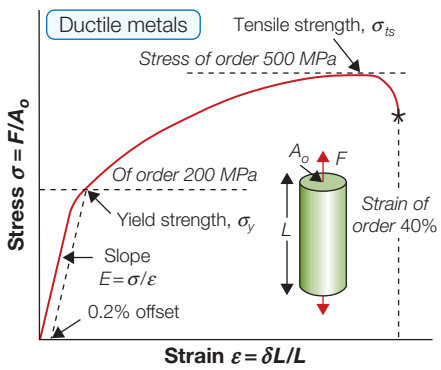


FIGURE 3.3
The stress-strain curve for a metal, showing the modulus, E, the 0.2% yield strength, σ_y, and the ultimate strength, σ_{ts}.

Mechanical properties

The *elastic modulus*, E (units: GPa or GN/m^2), is the slope of the initial, linear-elastic, part of the stress-strain curve (Figure 3.3). Young's modulus, E, describes response to tensile or compressive loading; the shear modulus, G, describes response to shear loading; and the bulk modulus, K, describes the response to hydrostatic pressure. Poisson's ratio, ν, is the negative of the ratio of the lateral strain, ε_2, to the axial strain, ε_1, in axial loading:

$$\nu = -\frac{\varepsilon_2}{\varepsilon_1}$$

In reality, moduli measured as slopes of stress-strain curves are inaccurate, often low by a factor of 2 or more, because of contributions to the strain from inelasticity, creep, and other factors. Accurate moduli are measured dynamically: by exciting the natural vibrations of a beam or a wire or by measuring the velocity of sound waves in the material.

In an isotropic material, the moduli are related in the following ways:

$$E = \frac{3G}{1+G/3K} \quad G = \frac{E}{2(1+\nu)} \quad K = \frac{E}{3(1-2\nu)} \tag{3.1}$$

Commonly,

$$\nu \approx 1/3$$

when

$$G \approx \frac{3}{8}E$$

and

$$K \approx E \tag{3.2a}$$

Elastomers are exceptional. For these,

$$\nu \approx 1/2$$

when

$$G \approx \frac{1}{3}E$$

and

$$K >> E \tag{3.2b}$$

Data sources like those described in Appendix D list values for all four moduli. In this book we examine data for E; approximate values for the others can be derived from the (3.2) equations when needed.

Estimating moduli

Young's modulus E for copper is 124 GPa; its Poisson's ratio ν is 0.345. What is its shear modulus, G?

Answer

Inserting the values for E and ν in the central equation (3.1) gives G = 46.1 GPa. The measured value is 45.6 GPa, a difference of only 1%.

The *strength*, σ_f (units: MPa or MN/m^2), of a solid requires careful definition. For metals, we identify σ_f with the 0.2% offset yield strength σ_y (see Figure 3.3), that is, the stress at which the stress-strain curve for axial loading deviates by a strain of 0.2% from the linear-elastic line. It is the same in tension and compression. For polymers, σ_f is identified as the stress at which the stress-strain curve becomes markedly nonlinear, at a strain typically of 1% (Figure 3.4). This may be caused by shear yielding: the irreversible slipping of molecular chains; or it may be caused by crazing: the formation of low-density, crack-like volumes that scatter light, making the polymer look white. Polymers are a little stronger ($\approx$ 20%) in compression than in tension.

Strength, for ceramics and glasses, depends strongly on the mode of loading (Figure 3.5). In tension, "strength" means the fracture strength, σ_t. In compression it means the crushing strength σ_c, which is much greater; typically

$$\sigma_c = 10 \text{ to } 15\, \sigma_t \tag{3.3}$$

When a material is difficult to grip, as is a ceramic, its strength can be measured in bending. The *flexural strength* or *modulus of rupture*, σ_{flex} (units: MPa) is the maximum surface stress in a bent beam at the instant of failure (Figure 3.6). One might expect this to be the same as the strength measured in tension, but for ceramics it is greater by a factor of about 1.3 because the volume subjected to this maximum stress is small and the probability of a large flaw lying in it is small also; in simple tension all flaws see the same stress.

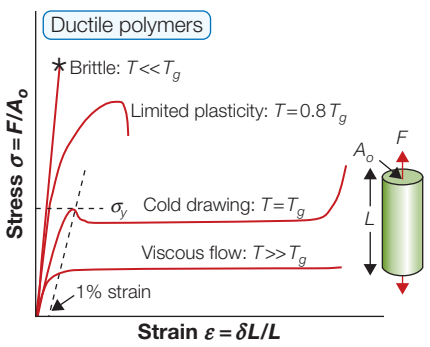


FIGURE 3.4
Stress-strain curves for a polymer below, at, and above its glass transition temperature, T_g.

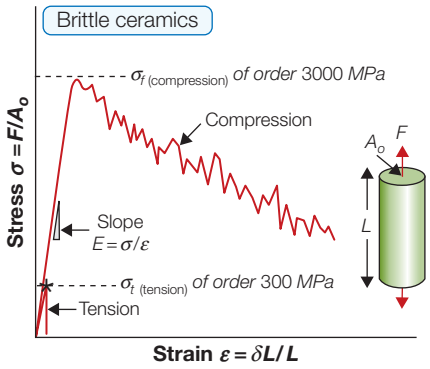


FIGURE 3.5
Stress-strain curves for a ceramic in tension and in compression. The compressive strength, σ_c, is 10 to 15 times greater than the tensile strength, σ_t.

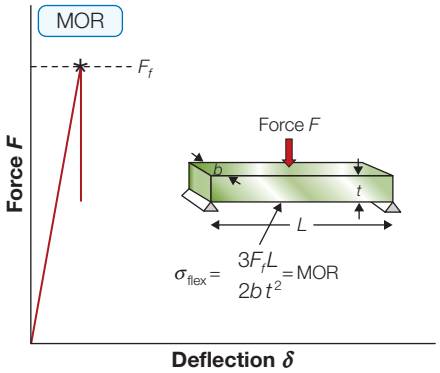


FIGURE 3.6
The modulus of rupture (MOR) is the surface stress at failure in bending. It is equal to, or slightly larger than, the failure stress in tension.

The strength of a composite is best defined by a set deviation from linear-elastic behavior; often an offset of 0.5% is taken. Composites that contain fibers, including natural composites such as wood, are a little weaker (up to 30%) in compression than tension because fibers buckle. In subsequent chapters, σ_f for composites means the tensile strength.

Strength, then, depends on material class and on mode of loading. Other modes of loading are possible: shear, for instance. Yield under multiaxial loads is related to that in simple tension by a *yield function*. For metals, the Von Mises yield function is a good description:

$$(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2 = 2\sigma_f{}^2 \tag{3.4}$$

where σ_1, σ_2, and σ_3 are the principal stresses, positive when tensile; σ_1, by convention, is the greatest or most positive; σ_3, the smallest or least positive. For polymers the yield function is modified to include the effect of pressure:

$$(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2 = 2\sigma_f{}^2\left(\frac{1+\beta p}{K}\right)^2 \tag{3.5}$$

where K is the bulk modulus of the polymer, $\beta \approx 2$ is a numerical coefficient that characterizes the pressure dependence of the flow strength, and the pressure p is defined by

$$p = -\frac{1}{3}(\sigma_1+\sigma_2+\sigma_3)$$

For ceramics, a Coulomb flow law is used:

$$\sigma_1 - B\sigma_2 = C \tag{3.6}$$

where B and C are constants.

The *tensile* (or *ultimate*) *strength* σ_{ts} (units: MPa) is the nominal stress at which a round bar of the material, loaded in tension, separates (Figure 3.3). For brittle solids—ceramics, glasses, and brittle polymers—it is the same as the failure strength in tension. For metals, ductile polymers, and most composites, it is greater than the yield strength, σ_y, by a factor of between

Using yield functions

A metal pipe of radius r and wall thickness t carry an internal pressure p. The pressure generates a circumferential wall stress of $\sigma_1 = pr/t$, an axial wall stress $\sigma_2 = pr/2t$. At what pressure will the pipe first yield?

Answer

Setting $\sigma_2 = \sigma_1/2$, $\sigma_3 = 0$ and $\sigma_f = \sigma_y$ in Equation (3.4) gives the yield condition $\sigma_1 = (2/\sqrt{3})\sigma_y$. Thus the pressure p^* that just causes first yield is $p^* = \frac{2}{\sqrt{3}}\frac{t}{r}\sigma_y$.

1.1 and 3 because of work hardening or, in the case of composites, load transfer to the reinforcement.

Cyclic loading can cause a crack to nucleate and grow in a material, culminating in fatigue failure. For many materials there exists a fatigue or *endurance limit*, σ_e (units: MPa), illustrated by the $\Delta\sigma - N_f$ curve of Figure 3.7. It is the stress amplitude $\Delta\sigma$ below which fracture does not occur, or occurs only after a very large number ($N_f > 10^7$) of cycles.

Tensile and compression tests are not always convenient: A large sample is needed and the test destroys it. The hardness test gives an approximate, nondestructive, measure of the strength. The *hardness*, H (SI units: MPa) of a material is measured by pressing a pointed diamond or hardened steel ball into the material's surface (Figure 3.8). The hardness is defined as the indenter force divided by the projected area of the indent. It is related to the quantity we have defined as σ_f by

$$H \approx 3\sigma_f \tag{3.7}$$

This, in the SI system, has units of MPa. Hardness is commonly reported in a bewildering array of other units, the most common of which is the Vickers H_v scale with units of kg/mm^2. It is related to H in the units used here by

$$H_v = \frac{H}{10}$$

A conversion chart for five hardness scales, relating them to yield strength, appears in Figure 3.9.

The *toughness*, G_{1c} (units: kJ/m^2), and the *fracture toughness*, K_{1c} (units: MPa/m$^{1/2}$ or MN/m$^{1/2}$), measure the resistance of a material to the propagation of a crack. The fracture toughness is measured by loading

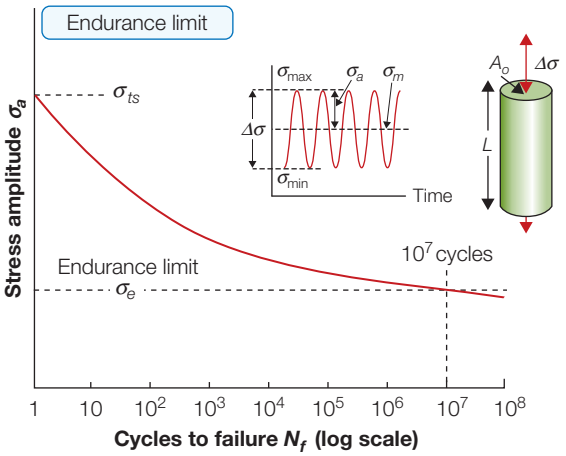


FIGURE 3.7
The endurance limit, σ_e, is the cyclic stress that causes failure in $N_f = 10^7$ cycles.

a sample containing a deliberately introduced crack of length $2c$ (Figure 3.10), recording the tensile stress σ^* at which the crack propagates. The quantity K_{1c} is then calculated from

$$K_{1c} = Y\sigma^*\sqrt{\pi c} \qquad (3.8)$$

and the toughness from

$$G_{1c} = \frac{K_{1c}^2}{E(1+\nu)} \qquad (3.9)$$

where Y is a geometric factor, near unity, that depends on details of the

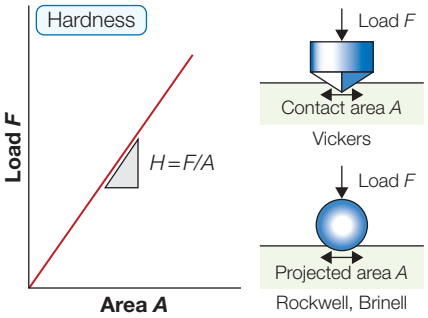


FIGURE 3.8
Hardness is measured as the load, F, divided by the projected area of contact, A, when a diamond-shaped indenter is forced into the surface.

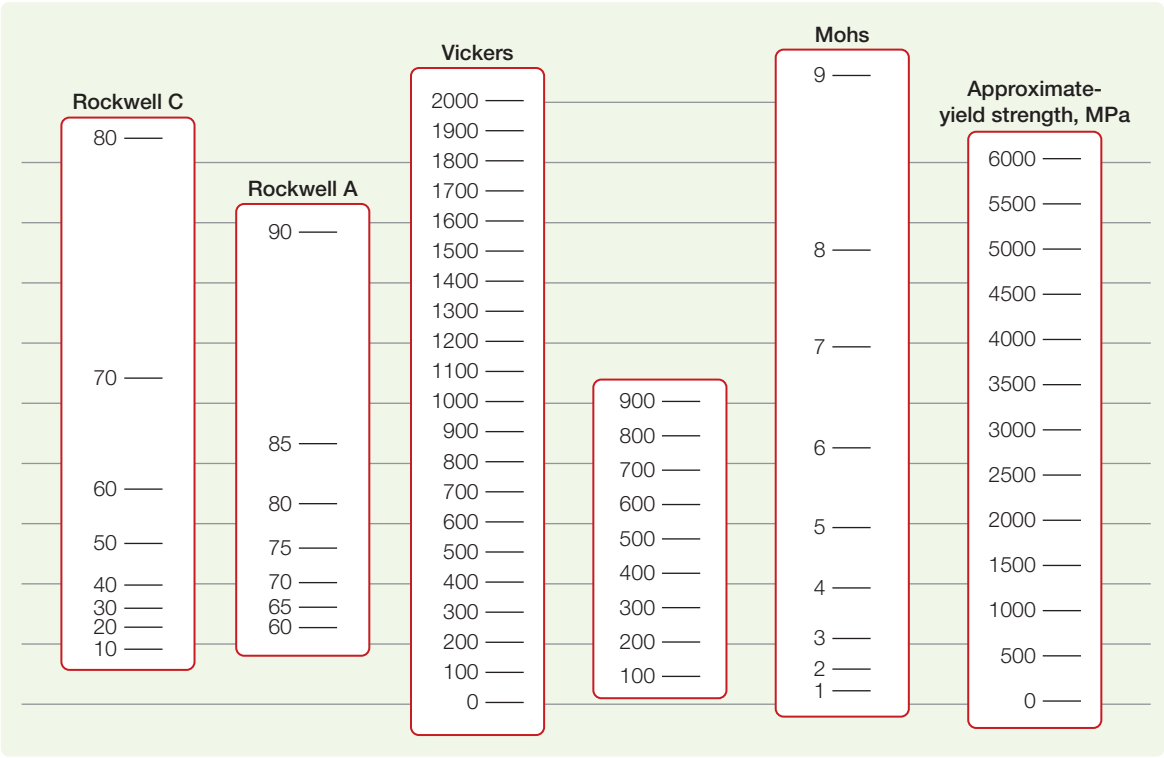


FIGURE 3.9
Commonly used scales of hardness related to each other and to the yield strength.

Strength from hardness

A steel has a hardness of 50 on the Rockwell C scale. Approximately what is its Vickers hardness and yield strength?

Answer

The chart of Figure 3.9 shows that the Vickers hardness corresponding to a Rockwell C value of 50 is approximately $H_v = 500$ and the yield strength is approximately 1,700 MPa.

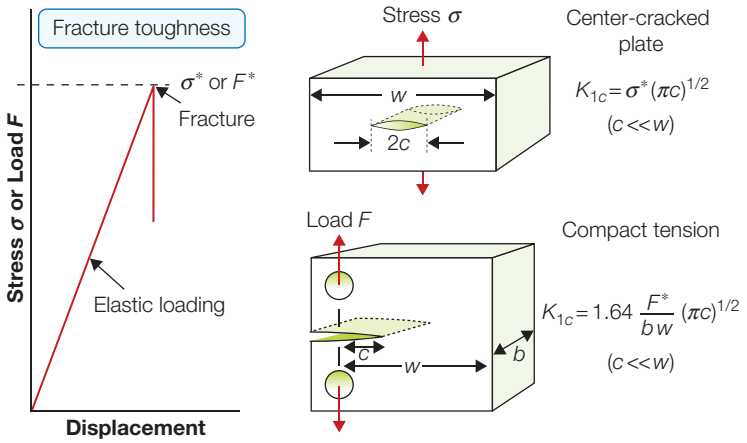


FIGURE 3.10

The fracture toughness, K_{1c}, measures the resistance to the propagation of a crack. The test specimen containing a crack of length $2c$ fails at stress σ^*. The fracture toughness is then $K_{1c} = Y\sigma^*\sqrt{\pi c}$ where Y is a constant near unity.

sample geometry, E is Young's modulus, and ν is Poisson's ratio. Measured in this way K_{1c} and G_{1c} have well-defined values for brittle materials (ceramics, glasses, and many polymers). In ductile materials a plastic zone develops at the crack tip, introducing new features into the way in which cracks propagate that necessitate more involved characterization. Values for K_{1c} and G_{1c} are, nonetheless, cited, and are useful as a way of ranking materials.

The *loss coefficient*, η (a dimensionless quantity), measures the degree to which a material dissipates vibrational energy (Figure 3.11). If a material is loaded elastically to a stress, σ_{max}, it stores an elastic energy

$$U = \oint_0^{\sigma_{max}} \sigma d\varepsilon \approx \frac{1}{2}\frac{\sigma_{max}^2}{E}$$

Using fracture toughness

A glass floor panel contains micro-cracks up to 2 microns in length. Glass has a fracture toughness of $K_{1c} = 0.6$ MPa.m$^{1/2}$. When the panel is walked upon, stresses as high as 30 MPa appear in it. Is it safe?

Answer

The stress required to make a 2-micron crack (so $c = 10^{-6}$ m) propagate in glass with a fracture toughness of $K_{1c} = 0.6$ MPa.m$^{1/2}$, using Equation 3.8 with $Y = 1$, is

$$\sigma_c = K_{1c}/\sqrt{\pi c} = 339 \text{ MPa}$$

The panel is safe.

per unit volume. If it is loaded and then unloaded, it dissipates an energy:

$$\Delta U = \oint \sigma d\varepsilon$$

The loss coefficient is

$$\eta = \frac{\Delta U}{2\pi U_{\max}} \qquad (3.10)$$

where $U_{\max}$ is the stored elastic energy at peak stress. The value of η usually depends on the time scale or frequency of cycling.

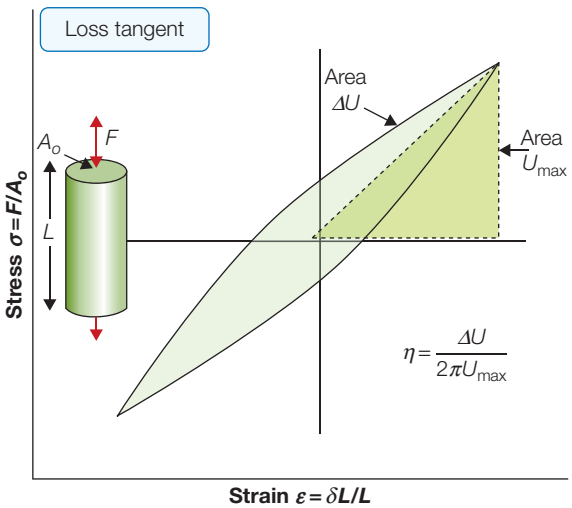


FIGURE 3.11
The loss tangent η measures the fractional energy dissipated in a stress-strain cycle.

Other measures of damping include the *specific damping capacity*, $D = \Delta U/U$, the *log decrement*, Δ (the log of the ratio of successive amplitudes of natural vibrations), the *phase lag*, δ, between stress and strain, and the "*Q*" *factor* or *resonance factor*, Q. When damping is small ($\eta < 0.01$) these measures are related by

$$\eta = \frac{D}{2\pi} = \frac{\Delta}{\pi} = \tan\delta = \frac{1}{Q} \qquad (3.11)$$

but when damping is large, they are no longer equivalent.

Wear, the loss of material when surfaces slide against each other, is a multibody problem. Nevertheless, it can, to a degree, be quantified. When solids slide (Figure 3.12), the volume of material lost from one surface, per unit distance slid, is called the wear rate, W (units: m^2). The wear resistance of the

Using loss coefficients

A bell with a natural frequency of $f = 1{,}000$ Hz is made of a material with a loss coefficient of $\eta = 0.01$. For how long will it ring after being struck? If the material is replaced by one of low damping with $\eta = 10^{-4}$ how long will it ring? (Assume the ring has ended when the amplitude of oscillation A has fallen to one-hundredth of its initial value.)

Answer

Let A and $A + dA$ be the amplitudes of the successive cycles (dA is negative). Then $\text{Log} = \left(\frac{A}{A + dA}\right) = \Delta = \pi\eta$ from which

$$\frac{dA}{Ad\eta} = \frac{1}{10^{\pi\eta}} - 1$$

Integrating over n cycles gives $\ln\frac{A}{A_0} = \left(\frac{1}{10^{\pi\eta}} - 1\right)n$ where A_0 is the initial amplitude. When A has fallen to 0.01 A_0, the term $\ln(A/A_0) = -4.6$, giving $n = 4.6\left(\frac{10^{\pi\eta}}{10^{\pi\eta} - 1}\right)$. Thus a bell with $\eta = 0.01$ will ring for $n = 66$ cycles, giving a time $n/f = 66$ milliseconds. A bell with $\eta = 10^{-4}$ will ring for $n = 6{,}400$ cycles and a time of n/f 6.4 seconds.

surface is characterized by the *Archard wear constant*, K_A (units: 1/MPa) defined by the equation

$$\frac{W}{A} = K_A P \tag{3.12}$$

where A is the area of the slider surface and P is the normal force pressing it onto the other surface. Approximate data for K_A appear in Chapter 4, but must be interpreted as the property of the sliding couple, not of just one member of it.

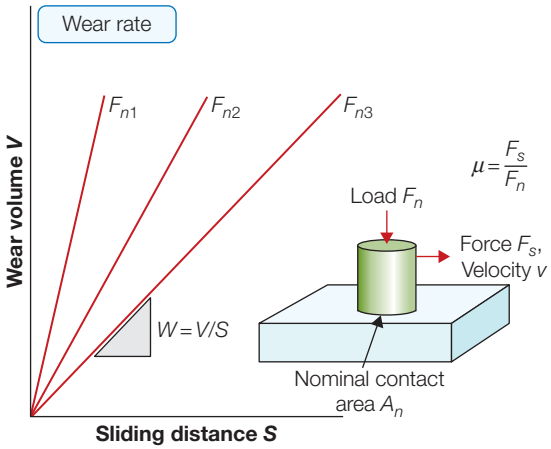


FIGURE 3.12
Wear is the loss of material from surfaces when they slide. The wear resistance is measured by the Archard wear constant, K_A, defined in the text.

Thermal properties

Two temperatures, the *melting temperature*, T_m, and the *glass temperature*, T_g (units for both: K or C), are fundamental because they relate directly to the strength of the bonds in the solid. Crystalline solids have a sharp melting point, T_m. Noncrystalline solids do not; the temperature T_g characterizes the transition from true solid to very viscous liquid. It is helpful, in engineering design, to define two further temperatures: the *maximum* and *minimum service temperature*, $T_{\max}$ and $T_{\min}$ (both: K or C). The first tells us the highest temperature at which the material can reasonably be used without oxidation, chemical change,

Calculating wear

A steel slider oscillates on a dry steel substrate at frequency $f = 0.2$ Hz and an amplitude $a = 2$ mm under a normal pressure $P = 2$ MPa. The Archard wear constant for steel on steel is $K_A = 3 \times 10^{-8}$ (MPa)$^{-1}$. By how much will the surface of the slider have been reduced in thickness after a time $t = 100$ hours?

Answer

The distance slid in 100 hours is $d = 4\ a\ f\ t$ m. The thickness x removed from the slider over the time $t = 3.6 \times 10^5$ is

$$x = \frac{\textit{Volume removed}}{\textit{Area } A} = 4\,a\,f\,t\,K_A\,P = 3.5 \times 10^{-5}\text{m} = 36\,\mu\text{m}$$

or excessive creep becoming a problem. The second is the temperature below which the material becomes brittle or otherwise unsafe to use.

It costs energy to heat a material. The *heat capacity* or *specific heat* (units J/kg.K) is the energy to heat 1 kg of a material by 1 K. The measurement is usually made at constant pressure (atmospheric pressure) so it is given the symbol C_p. When dealing with gases, it is more usual to measure the heat capacity at constant volume (symbol C_v), and for gases this differs from C_p. For solids the difference is so slight that it can be ignored, and we shall do so here. The heat capacity is measured by calorimetry (Figure 3.13), which is also the standard way of measuring the glass temperature, T_g. A measured quantity of energy (here, electrical energy) is pumped into a sample of material of known mass. The temperature rise is measured, allowing the energy/kg.K to be calculated. Real calorimeters are more elaborate than this, but the principle is the same.

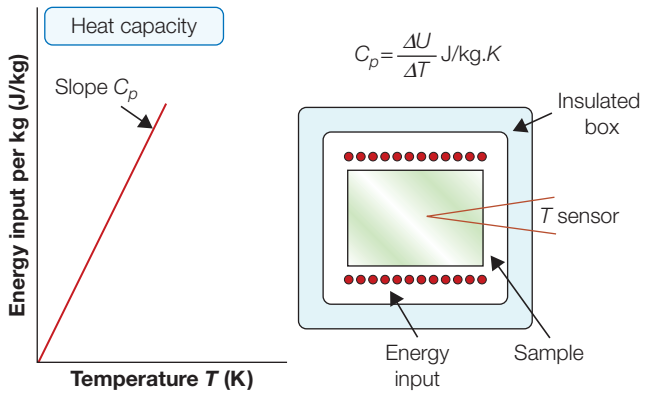


FIGURE 3.13

The heat capacity—the energy to raise the temperature of 1 kg of material by 1°C.

Using specific heat

How much energy is required to heat a 100 mm cube of copper from room temperature (20°C) to its melting point?

Answer

Data for melting point, T_m, specific heat, C_p, and density, ρ, are listed in Appendix A. The values for copper are $T_m = 1{,}082$°C, $C_p = 380$ J/kg.K and $\rho = 8{,}930$ kg/m^3. The mass of copper in the cube is $\rho V = 8.93$ kg. The energy to heat it to $\Delta T = 1{,}062$°C is

$$\rho V C_p \Delta T = 3.6 \text{ MJ}$$

(The energy in a liter of gasoline is 35 MJ.)

The rate at which heat is conducted through a solid at steady state (meaning that the temperature profile does not change with time) is measured by the *thermal conductivity*, λ (units: W/m.K). Figure 3.14 shows how it is measured: by recording the heat flux q (W/m^2) flowing through the material from a surface at higher temperature T_1 to a lower one T_2 separated by a distance X. The conductivity is calculated from Fourier's law:

$$q = -\lambda \frac{dT}{dX} = \lambda \frac{(T_1 - T_2)}{X} \tag{3.13}$$

The measurement is not, in practice, easy (particularly for materials with low conductivities), but reliable data are now generally available.

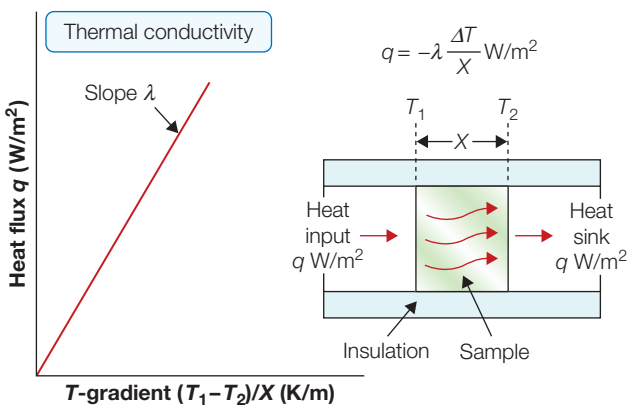


FIGURE 3.14

The thermal conductivity λ measures the flux of heat driven by a temperature gradient dT/dX.

Steady state heat flow

A heat exchanger has an exchange area of $A = 0.5\ \text{m}^2$. It passes heat from a fluid at temperature $T_1 = 100°\text{C}$ to a second fluid at $T_2 = 20°\text{C}$. The exchange wall is made of copper sheet (thermal conductivity $\lambda = 350$ W/m.K) with a thickness $X = 2$ mm. How much energy flows from one fluid to the other in one hour?

Answer

The temperature gradient $dT/dX = 80/0.002 = 40{,}000°\text{C}$ /m. The total energy Q that passes across the area A over a time $t = 3{,}600$ seconds is

$$Q = A\,t\,q = A\,t\,\lambda\,\frac{dT}{dX} = 2.5 \times 10^{10}\,\text{J} = 25\,\text{GJ}$$

When heat flow is transient, the flux depends instead on the *thermal diffusivity*, a (units: m^2/s), defined by

$$a = \frac{\lambda}{\rho\,C_p} \tag{3.14}$$

where ρ is the density and C_p is the heat capacity. The thermal diffusivity can be measured directly by measuring the decay of a temperature pulse when a heat source, applied to the material, is switched off; or it can be calculated from λ, via the last equation. The distance x heat diffuses in a time t is approximately

$$x \approx \sqrt{2\,\alpha\,t} \tag{3.15}$$

Most materials expand when they are heated (Figure 3.15). The thermal strain per degree of temperature change is measured by the *linear thermal-expansion coefficient*, α (units: K^{-1} or, more conveniently, "microstrain/°C" or $10^{-6}°\text{C}^{-1}$). If the material is thermally isotropic, the volume expansion,

Transient heat flow

You pour boiling water into a tea-glass with a wall thickness $x = 3$ mm. How many seconds have you got to carry it to the table before it becomes to hot to hold? (The thermal conductivity of glass is $\lambda = 1.1$ W/m.K, its density is $\rho = 2{,}450\ \text{kg/m}^3$ and its heat capacity $C_p = 800$ J/kg.K.)

Answer

Inserting the data into Equation (3.14) gives a thermal diffusivity for glass of $a = 5.6 \times 10^{-7}\ \text{m}^2/\text{s}$. Inserting this into Equation (3.13) gives the approximate time

$$t \approx \frac{x^2}{2a} = 8\,\text{seconds}$$

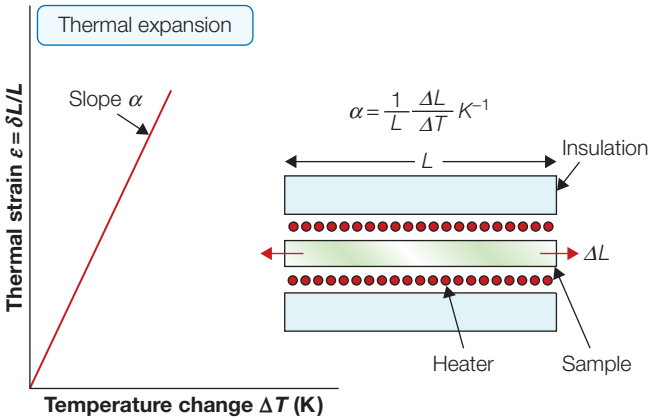


FIGURE 3.15
The linear-thermal expansion coefficient α measures the change in length, per unit length, when the sample is heated.

Thermal stress

An aluminum pipe is rigidly clamped to the face of a concrete building. On a hot day the face of the building in direct sun rises to 80°C, and because the expansion of the aluminum is greater than that of concrete, stress appears in it. What is the value of the stress if the original clamping was done on a day when the temperature was 20°C?

Answer

The expansion coefficient of aluminum is $\alpha = 22.5 \times 10^{-6}$/°C, that of concrete is $\alpha = 9 \times 10^{-6}$/°C, using means of the ranges in Appendix A. The aluminum pipe is rigidly clamped, so the difference in thermal strain $\Delta\alpha\Delta T = 13.5 \times 10^{-6} \times 60 = 8.1 \times 10^{-4}$. This has to be accommodated by elastic compression of the aluminum (modulus $E = 75$ GPa from Appendix A), giving a stress $\Delta\alpha\Delta T E = 61$ MPa. This is enough to cause a soft aluminum to yield.

per degree, is 3α. If it is anisotropic, two or more coefficients are required, and the volume expansion becomes the sum of the principal thermal strains.

The *thermal shock resistance* ΔT_s (units: K or C) is the maximum temperature difference through which a material can be quenched suddenly without damage. I, and the *creep resistance* are important in high-temperature design. Creep is the slow, time-dependent deformation that occurs when materials are loaded above about $\frac{1}{3}T_m$ or $\frac{2}{3}T_g$. Design against creep is a specialized subject. Here we rely instead on avoiding the use of a material above its maximum service temperature, $T_{\max}$, or , for polymers, the "heat deflection temperature."

Electrical properties

The *electrical resistivity*, ρ_e (SI units Ω.m or, commonly, $\mu\Omega$.cm), is the resistance of a unit cube with unit potential difference between a pair of its faces (Figure 3.16). It has an immense range, from a little more than 10^{-8} in units of Ω.m for good conductors (equivalent to 1 $\mu\Omega$.cm) to more than 10^{16} Ω.m (10^{24} $\mu\Omega$.cm) for the best insulators. The electrical conductivity, κ_e (units Siemens per meter, S/m or $(\Omega.m)^{-1}$), is simply the reciprocal of the resistivity.

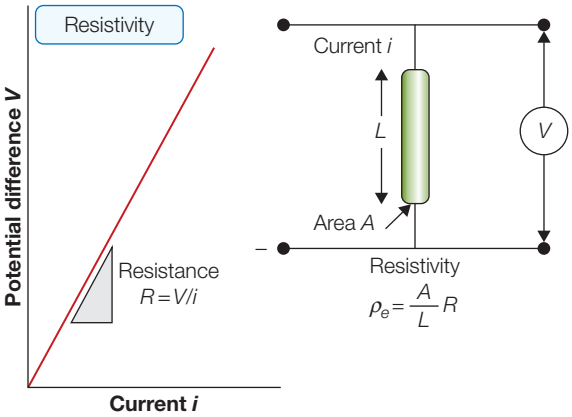


FIGURE 3.16
Electrical resistivity, ρ_e, is measured as the potential gradient, V/L, divided by the current density, i/A. It is related to resistance, R, by $\rho_e = AR/L$.

When an insulator (or *dielectric*) is placed in an electric field, it becomes polarized and charges appear on its surfaces that tend to screen the interior from the electric field. The tendency to polarize is measured by the *dielectric constant*, ε_r, a dimensionless quantity (Figure 3.17). Its value for free space and, for practical purposes, for most gasses, is 1. Most insulators have values between 2 and 30, though low-density foams approach the value 1 because they are largely air.

What does ε_r measure? Two conducting plates separated by a dielectric make a capacitor. Capacitors store charge. The charge, Q (units: coulombs), is directly proportional to the voltage difference between the plates, V (volts):

$$Q = CV \tag{3.16}$$

where C (farads) is the capacitance. The capacitance of a parallel plate capacitor of area A, separated by empty space (or by air), is

$$C = \varepsilon_o \frac{A}{t} \tag{3.17}$$

Resistivity and resistance

Tungsten has a conductivity of $\kappa_e = 8.3 \times 10^6$ Siemens. What is the resistance of a tungsten wire of radius $r = 100$ microns in diameter and length $L = 1$ m?

Answer

The resistivity of tungsten $\rho_e = 1/\kappa_e = 1.2 \times 10^{-7}$ Ω.m. The resistance R of the wire is

$$R = \rho_e \frac{L}{\pi r^2} = 3.8\,\Omega$$

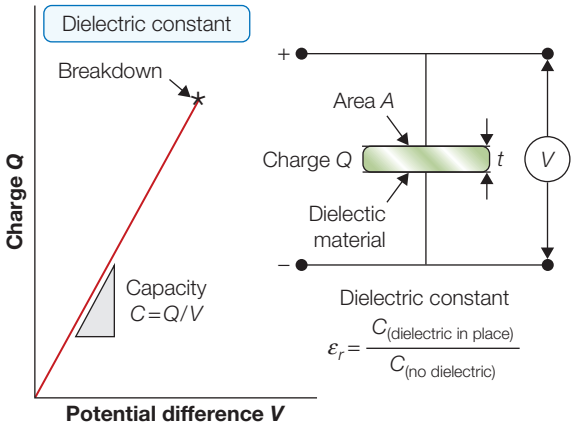


FIGURE 3.17
Dielectric constant: a measure of the ability of an insulator to polarize.

where ε_0 is the *permittivity of free* space (8.85×10^{-12} *F/m*, where *F* is farads). If the empty space is replaced by a dielectric, capacitance increases because of its polarization. The field created by the polarization opposes the field *E*, reducing the voltage difference *V* needed to support the charge. Thus the capacity of the condenser is increased to the new value:

$$C = \varepsilon \frac{A}{t} \tag{3.18}$$

where ε is the *permittivity of the dielectric* with the same units as ε_0. It is usual to cite not this but the *relative permittivity* or *dielectric constant*, ε_r:

$$\varepsilon_r = \frac{C_{with\ dielectric}}{C_{no\ dielectric}} = \frac{\varepsilon}{\varepsilon_o} \tag{3.19}$$

making the capacitance

$$C = \varepsilon_r \varepsilon_o \frac{A}{t} \tag{3.20}$$

When charged, the energy stored in a capacitor is

$$\frac{1}{2} Q V = \frac{1}{2} C V^2 \tag{3.21}$$

and this can be large: "Super-capacitors" with capacitances measured in farads store enough energy to power a hybrid car.

Polarization involves the small displacement of charge (of either electrons or ions) or of molecules that carry a dipole moment when an electric field is applied to the material. An oscillating field drives the charge between

Stray capacitance

The time constant τ for charging or discharging a capacitor is

$$\tau = R C$$

where *R* is the resistance of the circuit. That means that stray capacitance in an electronic circuit (capacitance between neighboring conducting lines or components) slows its response. What material choices minimize this?

Answer
Choosing materials with low resistivity ρ_e for the conductors (to minimize *R*) and choosing insulators with low dielectric constant ε_r to separate them (to minimize *C*), minimizes τ.

two alternative configurations. This charge motion is like an electric current that—if there were no losses—would be 90° out of phase with the voltage. In real dielectrics this current dissipates energy, just as a current in a resistor does, giving it a small phase shift, δ (Figure 3.18). The *loss tangent*, *tan* δ, also called the *dissipation factor*, D, is the tangent of the loss angle. The *power factor*, P_f, is the sine of the loss angle. When δ is small, as it is for the materials of interest here, all three are essentially equivalent:

$$P_f \approx D \approx \tan\delta \approx \sin\delta \qquad (3.22)$$

More useful, for our purposes, is the *loss factor* L, which is the loss tangent times the dielectric constant:

$$L = \varepsilon_r \tan\delta \qquad (3.23)$$

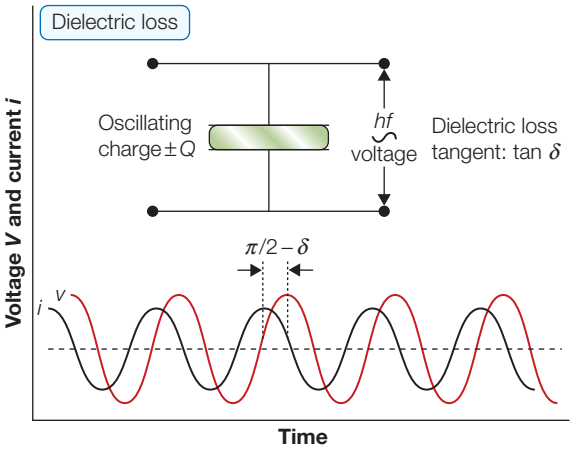


FIGURE 3.18
Dielectric loss, important in dielectric heating, as explained in the text.

It measures the energy dissipated by a dielectric when in an oscillating field. If you want to select materials to minimize or maximize dielectric loss, then the measurement you want is L.

When a dielectric material is placed in a cyclic electric field of amplitude E and frequency f, power P is dissipated and the field is correspondingly attenuated. The power dissipated per unit volume (W/m^3) is

$$P \approx f E^2 \varepsilon \tan\delta = f E^2 \varepsilon_o \varepsilon_r \tan\delta = f E^2 \varepsilon_o L \qquad (3.24)$$

Dielectric heating

A nylon component is placed in a microwave cavity with a field strength $E = 10^4$ V/m and a frequency $f = 10^{10}$ Hz for a time $t = 100$ s. The dielectric loss factor for nylon is $L = 0.1$, its density is $\rho = 1130$ kg/m^3 and its heat capacity is $C_p = 1650$ J/kg.K. Assuming no heat loss, how hot will the component become?

Answer

The heat generated by the field is $Q = Pt = f E^2 \varepsilon_0 L\, t = 8.85 \times 10^7$ J/m^3. The heat capacity of nylon per unit volume is $C_p \rho = 1.86 \times 10^6$ J/m^3.K. The temperature rise, ΔT is

$$\Delta T = \frac{f E^2 \varepsilon_o L\, t}{C_p \rho} = 47.6\,^\circ\text{C}$$

where, as before, ε_r is the dielectric constant of the material and *tan* δ is its loss tangent. This power appears as heat; the higher the frequency or the field strength and the greater the loss factor $L = \varepsilon_r \tan \delta$, the greater the heating and energy loss. Sometimes this dielectric loss is exploited in processing—for example, in radio frequency welding of polymers.

The *breakdown potential* (units: MV/m) is the electrical potential gradient at which an insulator breaks down and a damaging surge of current flows through it. It is measured by increasing, at a uniform rate, a 60-Hz alternating potential applied across the faces of a plate of the material until breakdown occurs, typically at a potential gradient between 1 and 100 million volts per meter (units: MV/m).

Optical properties

All materials allow for some passage of light, although for metals it is exceedingly small. The speed of light when in the material, v, is always less than that in vacuum, c. A consequence is that a beam of light striking the surface of such a material at an angle of incidence, α, enters the material at an angle β, the angle of refraction. The *refractive index*, n (dimensionless), is

$$n = \frac{c}{v} = \frac{\sin \alpha}{\sin \beta} \tag{3.25}$$

It is related to the dielectric constant, ε_r, at the same frequency by

$$n \approx \sqrt{\varepsilon_r}$$

The refractive index depends on wavelength and thus on the color of the light. The denser the material, and the higher its dielectric constant, the greater the refractive index. When $n = 1$, the entire incident intensity enters the material, but when $n > 1$, some is reflected. If the surface is smooth and polished, it is reflected as a beam; if rough, it is scattered. The percentage reflected, R, is related to the refractive index by

$$R = \left(\frac{n-1}{n+1}\right)^2 \times 100 \tag{3.26}$$

As n increases, the value of R approaches 100%.

Eco-properties

The *embodied energy* (units MJ/kg) is the energy required to extract 1 kg of a material from its ores and feedstock. The associated CO_2 *footprint* (units: kg/kg) is the mass of carbon dioxide released into the atmosphere during the production of 1 kg of material. These and other eco-attributes are the subject of Chapter 15.

3.5 SUMMARY AND CONCLUSIONS

There are six important families of materials for mechanical design: metals, ceramics, glasses, polymers, elastomers, and hybrids (which combine the properties of two or more of the others). Within a family there is a certain commonality. Ceramics and glasses as a family are hard, brittle, and corrosion-resistant. Metals are ductile, tough and good thermal and electrical conductors. Polymers are light, easily shaped, and electrical insulators. Elastomers have the ability to deform elastically to large strains. That is what makes the classification useful. But in design we wish to escape from the constraints of family and think, instead, of the material name as an identifier for a certain property profile—one that will in later chapters be compared with an "ideal" profile suggested by the design, guiding our choice. To that end, the properties important in thermo-mechanical design were defined in this chapter. In the next chapter we develop a way of displaying these properties so as to maximize the freedom of choice.

3.6 FURTHER READING

Definitions of material properties can be found in numerous general texts on engineering materials, among them those listed here.

Ashby, M.F., & Jones, D.R.H. (1996). *Engineering materials 1, an introduction to properties, applications and design* (3rd ed.). Elsevier–Butterworth-Heinemann, ISBN 0-7506-6380-4.
An introduction to materials, taking a design-led approach.

Ashby, M.F., Shercliff, H.R., & Cebon, D. (2010). *Materials: engineering, science, processing and design* (2nd ed.). Butterworth-Heinemann, ISBN 978-1-85617-895-2.
An elementary text introducing materials through material property charts and developing selection methods through case studies.

Askeland, D.R., & Phulé, P.P. (2006). *The science and engineering of materials* (5th ed.). Thomson, ISBN 0-534-55396-6.
A widely used introductory text.

ASM Handbooks, Vol. 8 (2004). *Mechanical testing and evaluation.* ASM International.
An online, subscription-based resource, detailing testing procedures for metals and ceramics.

ASM Engineered Materials Handbook (2004). *Testing and characterization of polymeric materials.* ASM International.
An online, subscription-based resource detailing testing procedures for polymers.

ASTM Standards, Vol. 08.01 and *08.02* (1988). *Plastics, Vol. 04.02* (1989). *Concrete, Vols. 01.01* to *01.05* (1990). *Steels, Vol. 02.01. Copper alloys, Vol. 02.03. Aluminum alloys, Vol. 02.04. Nonferrous alloys, Vol. 02.05. Coatings, Vol. 03.01. Metals at high and low temperatures, Vol. 04.09. Wood, Vol. 09.01* and *09.02*. American Society for Testing Materials, ISBN 0-8031-1581-4.
The ASTM set standards for materials testing.

Budinski, K.G., & Budinski, M.K. (2010). Engineering materials, properties and selection (9th ed.). Prentice Hall, ISBN 978-0-13-712842-6.
A well-established materials text that deals well with both material properties and processes.

Callister, W.D. (2010). *Materials science and engineering, an introduction* (8th ed.). John Wiley, ISBN 978-0-470-41997-7.
A well-respected materials text, now in its 7th edition, widely used for materials teaching in North America.

Charles, J.A., Crane, F.A.A., & Furness, J.A.G. (1997). *Selection and use of engineering materials* (3rd ed.). Butterworth-Heinemann, ISBN 0-7506-3277-1.
A materials science approach to the selection of materials.

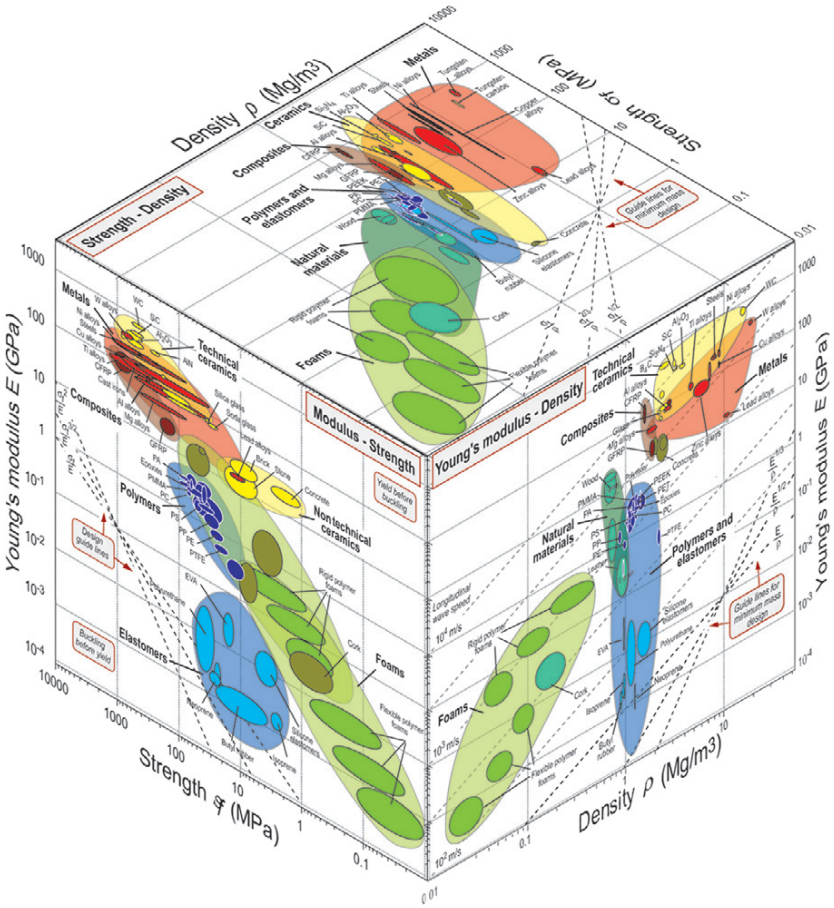
Dieter, G.E. (1999). *Engineering design, a materials and processing approach* (3rd ed.). McGraw-Hill, ISBN 9-780-073-66136-0.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

Farag, M.M. (2008). *Materials and process selection for engineering design* (2nd ed.). CRC Press, Taylor and Francis: ISBN 9-781-420-06308-0.
A materials science approach to the selection of materials.

Shackelford, J.F. (2009). *Introduction to materials science for engineers* (7th ed.). Prentice Hall, ISBN 978-0-13-601260-3.
A well-established materials text with a design slant.

CHAPTER 4

Material Property Charts



A 3-dimensional slice through material property space: the modulus–strength–density slice.

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4.1 INTRODUCTION AND SYNOPSIS

Material properties limit performance. We need a way of surveying them to get a feel for the values that design-limiting properties can have. One property can be displayed as a ranked list or bar chart, but it is seldom that the performance of a component depends on just one property. More often it is a combination of properties that matter: the need for stiffness at low weight, for thermal conduction coupled with corrosion resistance, or for strength combined with toughness, for example. This suggests the idea of plotting one property against another, mapping out the fields in property space occupied by each material class and the subfields occupied by individual materials.

The resulting charts are helpful in many ways. They condense a large body of information into a compact but accessible form; they reveal correlations between material properties that aid in checking and estimating data; and, as examined in later chapters, they become tools for materials selection, for exploring the effect of processing on properties, for demonstrating how shape can enhance structural efficiency, and for suggesting directions for further material development.

The ideas behind materials selection charts are described briefly in Section 4.2. Section 4.3 introduces the charts themselves. It's not necessary to read it all, but it is helpful to persist far enough to be able to read and interpret the charts fluently, and to understand the meaning of the design guide lines that appear in them. If, later, you use a particular chart, you should read the background to it, given here, to be sure of interpreting it correctly.

As explained in the Preface, the charts can be copied and distributed for teaching purposes without infringing the copyright.[1]

4.2 EXPLORING MATERIAL PROPERTIES

Each property of an engineering material has a characteristic range of values. The span can be large: many properties have values that range over five or more decades. One way of displaying this is as a bar chart like that of Figure 4.1 for Young's modulus. Each bar describes one material; its length shows the range of modulus exhibited by that material in its various forms. The materials are segregated by class. Each class shows a characteristic range: Metals and ceramics have high moduli; polymers have low; hybrids have a wide range, from low to high. The total range is large—it spans a factor of about 10^6—so logarithmic scales are used to display it.

More information is displayed by an alternative plot, illustrated in the schematic of Figure 4.2. Here, one property (the modulus, E, in this case) is plotted against another (the density, ρ). The range of the axes is chosen to include all materials, from the lightest, flimsiest foams to the stiffest, heaviest metals, and it is large, again requiring log scales. It is found that data for a given family of materials (polymers, for example) cluster together; the *subrange* associated with one material family is, in all cases, much smaller than the *full* range of that property. Data for one family can be enclosed in a property envelope—envelopes are shown on this schematic. A real $E - \rho$ chart is shown in Figure 4.3. The family envelopes appear as in the schematic. Within each envelope lie white bubbles enclosing classes and subclasses.

All of this is simple enough—just a helpful way of plotting data. But by choosing the axes and scales appropriately, more can be added. The speed

[1] A set of charts can be downloaded from *www.grantadesign.com*. All charts shown in this chapter were created using Granta Design's CES Edu Materials Selection software. With it you can make charts with any pair (or combination) of properties as axes.

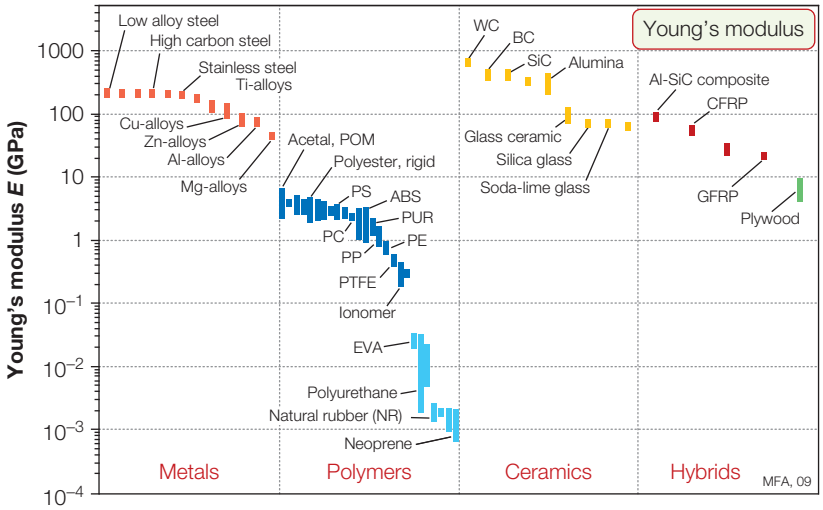


FIGURE 4.1

A bar chart showing modulus for families of solids. Each bar shows the range of modulus offered by a material, some of which are labeled.

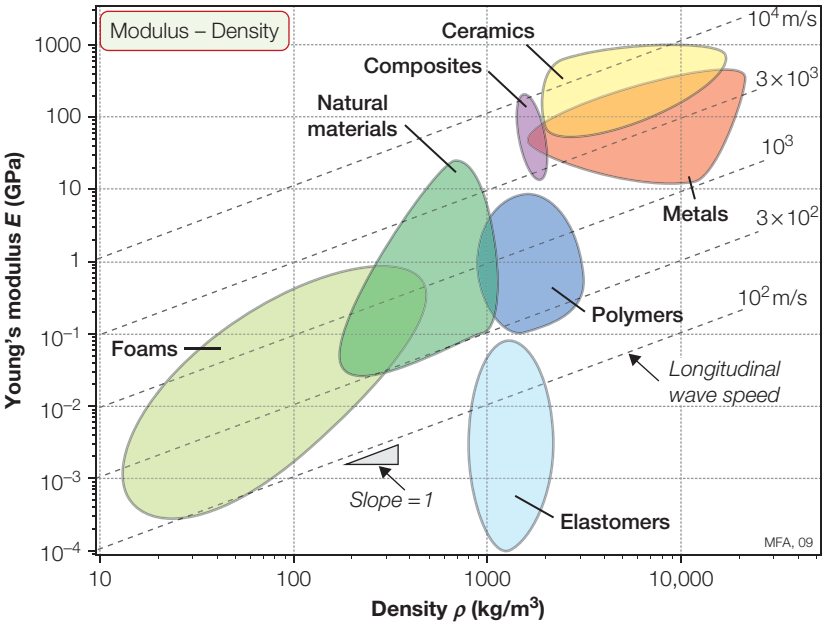


FIGURE 4.2

The idea of a material property chart: Young's modulus E is plotted against the density ρ on log scales. Each material class occupies a characteristic field. The contours show the longitudinal elastic wave speed $v = (E/\rho)^{1/2}$.

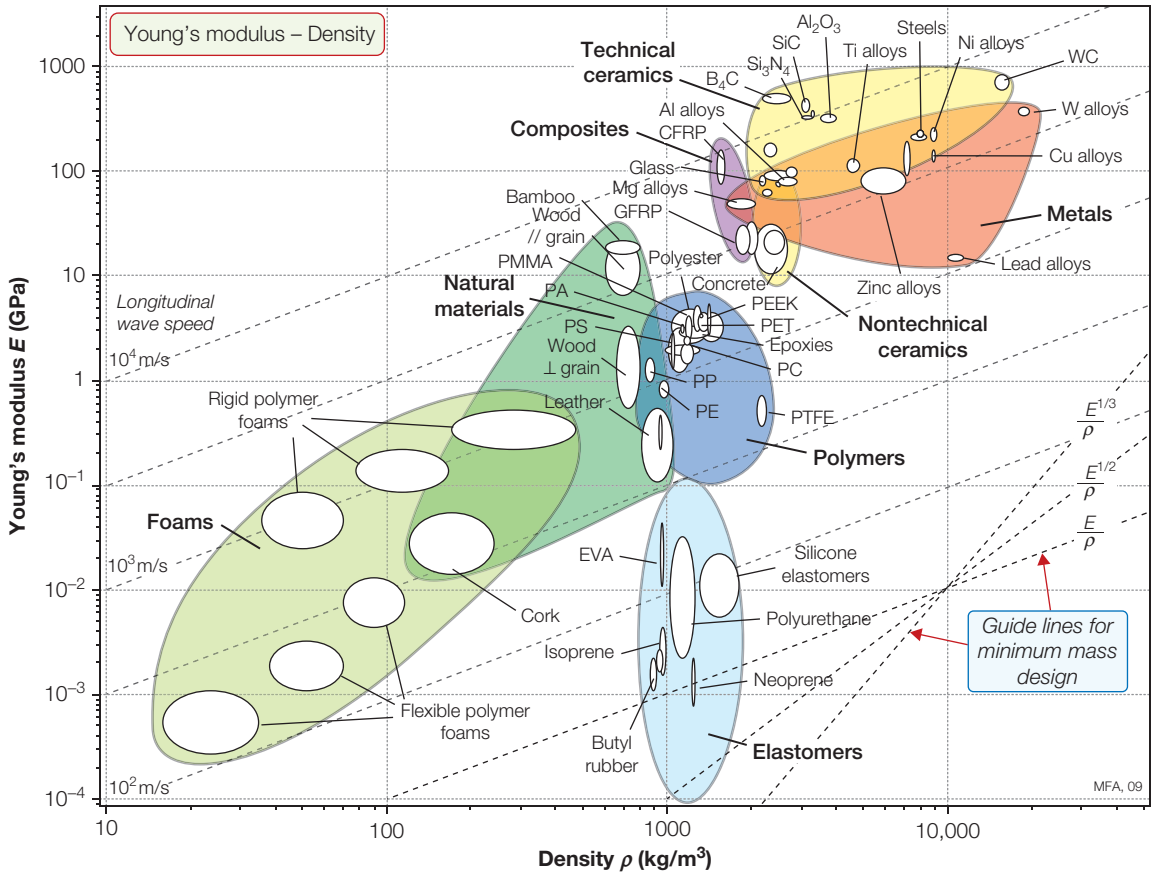


FIGURE 4.3

Young's modulus E plotted against density ρ. The heavy envelopes enclose data for a given class of material. The diagonal contours show the longitudinal wave velocity. The guide lines of constant E/ρ, $E^{1/2}/\rho$, and $E^{1/3}/\rho$ allow selection of materials for minimum weight, deflection-limited, design.

of sound in a solid depends on E and ρ; the longitudinal wave speed v, for instance, is

$$v = \left(\frac{E}{\rho}\right)^{1/2}$$

or (taking logs)

$$\log E = \log \rho + 2 \log v$$

For a fixed value of v, this equation plots as a straight line of slope 1 in Figures 4.2 and 4.3. This allows us to add *contours of constant wave velocity* to

the chart: They are the family of parallel diagonal lines, linking materials in which longitudinal waves travel with the same speed. All the charts allow additional fundamental relationships of this sort to be displayed. And there is more: Design-optimizing parameters called *material indices* also plot as contours on the charts (see Chapter 5).

Among the mechanical and thermal properties, there are 30 or so that are of primary importance, both in characterizing the material and in engineering design. They were listed in Table 3.1 and include density, moduli, strength, hardness, toughness, thermal and electrical conductivities, expansion coefficient, and specific heat. The charts display data for these properties for the families and classes of materials listed in Table 4.1. The list is expanded from the original six of Figure 3.1 by distinguishing *composites* from *foams* and from *natural materials,* and by distinguishing the high-strength *technical ceramics* (e.g., silicon carbide) from the low-strength, *nontechnical ceramics* (e.g., concrete and brick). Within each family, data are plotted for a

Table 4.1 Material Families and Classes

Family	Classes	Short Name
Metals (the metals and alloys of engineering)	Aluminum alloys	Al alloys
	Copper alloys	Cu alloys
	Lead alloys	Lead alloys
	Magnesium alloys	Mg alloys
	Nickel alloys	Ni alloys
	Carbon steels	Steels
	Stainless steels	Stainless steels
	Tin alloys	Tin alloys
	Titanium alloys	Ti alloys
	Tungsten alloys	W alloys
	Lead alloys	Pb alloys
	Zinc alloys	Zn alloys
Ceramics, *technical ceramics* (fine ceramics capable of load-bearing application)	Alumina	Al_2O_3
	Aluminum nitride	AlN
	Boron carbide	B_4C
	Silicon carbide	SiC
	Silicon nitride	Si_3N_4
	Tungsten carbide	WC
Ceramics, *nontechnical ceramics* (porous ceramics of construction)	Brick	Brick
	Concrete	Concrete
Glasses	Soda-lime glass	Soda-lime glass
	Borosilicate glass	Borosilicate glass
	Silica glass	Silica glass
	Glass ceramic	Glass ceramic

Table 4.1 *continued*

Family	Classes	Short Name
Polymers (the thermoplastics and thermosets of engineering)	Acrylonitrile butadiene styrene	ABS
	Cellulose polymers	CA
	Ionomers	Ionomers
	Epoxies	Epoxy
	Phenolics	Phenolics
	Polyamides (nylons)	PA
	Polycarbonate	PC
	Polyesters	Polyester
	Polyetheretherkeytone	PEEK
	Polyethylene	PE
	Polyethylene terephthalate	PET or PETE
	Polymethylmethacrylate	PMMA
	Polyoxymethylene (Acetal)	POM
	Polypropylene	PP
	Polystyrene	PS
	Polytetrafluorethylene	PTFE
	Polyvinylchloride	PVC
Elastomers (engineering rubbers, natural and synthetic)	Butyl rubber	Butyl rubber
	EVA	EVA
	Isoprene	Isoprene
	Natural rubber	Natural rubber
	Polychloroprene (Neoprene)	Neoprene
	Polyurethane	PU
	Silicone elastomers	Silicones
Hybrids: composites	Carbon-fiber–reinforced polymers	CFRP
	Glass-fiber–reinforced polymers	GFRP
	SiC-reinforced aluminum	Al-SiC
Hybrids: foams	Flexible polymer foams	Flexible foams
	Rigid polymer foams	Rigid foams
Hybrids: natural materials	Cork	Cork
	Bamboo	Bamboo
	Wood	Wood

representative set of materials, chosen both to span the full range of behavior for the class and to include the most common and most widely used members of it. In this way the envelope for a family encloses data not only for the materials listed in Table 4.1 but for virtually all other members of the family as well.

The charts that follow show a *range* of values for each property of each material. Sometimes the range is narrow: The modulus of copper, for instance, varies by only a few percent about its mean value, influenced by purity, texture, and the like. Sometimes, however, it is wide: The strength of metals can vary by a factor of 100 or more, influenced by composition and the state of work hardening or heat treatment. Crystallinity and degree of cross-linking greatly influence the modulus of polymers. Porosity influences the strength of ceramics. These *structure-sensitive* properties appear as elongated bubbles within the envelopes in the charts.

4.3 THE MATERIAL PROPERTY CHARTS

The modulus–density chart

Modulus and density are familiar properties. Steel is stiff; rubber is compliant: These are effects of modulus. Lead is heavy; cork is buoyant: these are effects of density. Figure 4.3 shows the range of Young's modulus, E, and density, ρ, for engineering materials. Data for members of a particular family of materials cluster together and can be enclosed by a colored envelope. The same family envelopes appear on all the diagrams: They correspond to the main headings in Table 4.1.

The *density* of a solid depends on the atomic weight of its atoms or ions, their size, and the way they are packed. The size of atoms does not vary much: Most have a volume within a factor of two of 2×10^{-29} m^3. Packing fractions do not vary much either—a factor of two more or less. Close packing gives a packing fraction of 0.74; open networks such as that of the diamond-cubic structure give about 0.34. The spread of density comes mainly from the spread of atomic weight, ranging from 1 for hydrogen to 238 for uranium. Metals are dense because they are made of heavy atoms, packed closely together; polymers have low densities because they are largely made of carbon (atomic weight: 12) and hydrogen (atomic weight: 1) in more open amorphous or crystalline packings. Ceramics, for the most part, have lower densities than metals because they contain light O, N or C atoms. Even the lightest atoms, packed in the most open way, give solids with a density of around 1000 kg/m^3, the same as that of water. Materials with lower densities than this are *foams*—materials made up of cells containing a large fraction of pore space.

The *moduli* of most materials depend on two factors: bond stiffness and the number of bonds per unit volume. A bond is like a spring, and, like a spring, it has a spring constant, S (units: N/m). Young's modulus, E, is roughly

$$E = \frac{S}{r_o} \tag{4.1}$$

where r_o is the "atom size" (r_o^3 is the mean atomic or ionic volume). The wide range of moduli is largely caused by the range of values of S. The covalent bond is stiff (S = 20–200 N/m); the metallic and the ionic a little less so (S = 15–100 N/m). Diamond has a very high modulus because the carbon atom is small, giving a high bond density, and its atoms are linked by strong covalent springs (S = 200 N/m). Metals have high moduli because close packing gives a high bond density and the bonds are strong, though not as strong as those of diamond. Polymers contain both strong diamond-like covalent bonds and weak hydrogen or Van der Waals bonds (S = 0.5–2 N/m). It is the weak bonds that stretch when the polymer is deformed, giving low moduli.

But even large atoms ($r_o = 3 \times 10^{-10}$ m) bonded with the weakest bonds (S = 0.5 N/m) have a modulus of roughly

$$E = \frac{0.5}{3 \times 10^{-10}} \approx 1 \text{ GPa} \tag{4.2}$$

This is the *lower limit* for true solids. The chart shows that many materials have moduli that are lower than this: They are either elastomers or foams. Elastomers have a low E because their weak secondary bonds have melted as their glass temperature, T_g, is below room temperature, leaving only the very weak "entropic" restoring force associated with tangled, long-chain molecules. Foams have low moduli because the cell walls bend easily when the material is loaded. More on this in Chapter 11.

The chart shows that the modulus of engineering materials spans seven decades,[2] from 0.0001 GPa (low-density foams) to 1,000 GPa (diamond). The density spans a factor of 2,000, from less than 0.01 to 20 Mg/m^3. Ceramics as a family are very stiff, metals a little less so—but none have a modulus less than 10 GPa. Polymers, by contrast, all cluster between 0.8 and 8 GPa.

The log scales allow more information to be displayed. As explained in the last section, the velocity of elastic waves in a material and the natural vibration frequencies of a component made of it are proportional to $(E/\rho)^{1/2}$. Contours of this quantity are plotted on the chart, labeled with the longitudinal wave speed. The speed varies from less than 50 m/s (soft elastomers) to a little more than 10^4 m/s (stiff ceramics). We note that aluminum and glass, because of their low densities, transmit waves quickly despite their low moduli. One might have expected the wave velocity in foams to be low because of the low modulus, but the low density almost compensates. That in wood, across the grain, is low; but along the grain, it is high—roughly the same as steel—a fact made use of in the design of musical instruments.

[2] Very-low-density foams and gels (which can be thought of as molecular-scale, fluid-filled foams) can have lower moduli than this. As an example, gelatin (as in Jell-O) has a modulus of about 10^{-5} GPa. Their strengths and fracture toughness, too, can be below the lower limit of the charts.

Surveying material properties

Which class of metallic alloy is the lightest? Which the heaviest? Which the stiffest? Which the least stiff?

Answer

A glance at Figure 4.3 reveals that the lightest class is magnesium alloys and the heaviest is tungsten alloys; tungsten alloys are also the stiffest, and lead alloys are the least stiff.

All the charts that appear in this and subsequent chapters can be used for this fast access to comparisons.

Comparing sound velocities

A metal is needed in which longitudinal waves travel at 300 m/s. Use Figure 4.3 to identify candidates.

Answer

The chart shows that zinc alloys, copper alloys, and tungsten alloys all have a longitudinal wave speed close to 300 m/s.

The chart helps in the common problem of material selection for applications in which mass must be minimized. Guide lines corresponding to three common geometries of loading were shown in Figure 4.3. Their use in selecting materials for stiffness-limited design at minimum weight is described in Chapters 5 and 6.

The strength–density chart

The modulus of a solid is a well-defined quantity with a sharp value. The strength is not. It is shown, plotted against density, ρ, in Figure 4.4.

The word "strength" needs definition (see also Section 3.3). For metals and polymers, it is the *yield strength*, but since the range of materials includes those that have been worked or hardened in some other way as well as those that have been softened by annealing, the range is large. For brittle ceramics, the strength plotted here is the *modulus of rupture*: the flexural strength. It is slightly greater than the tensile strength, but much less than the compression strength, which for ceramics is 10 to 15 times greater than the strength in tension. For elastomers, strength means the tensile *tear strength*. For composites, it is the *tensile failure strength* (the compressive strength can be less by up to 30% because of fiber buckling). We will use

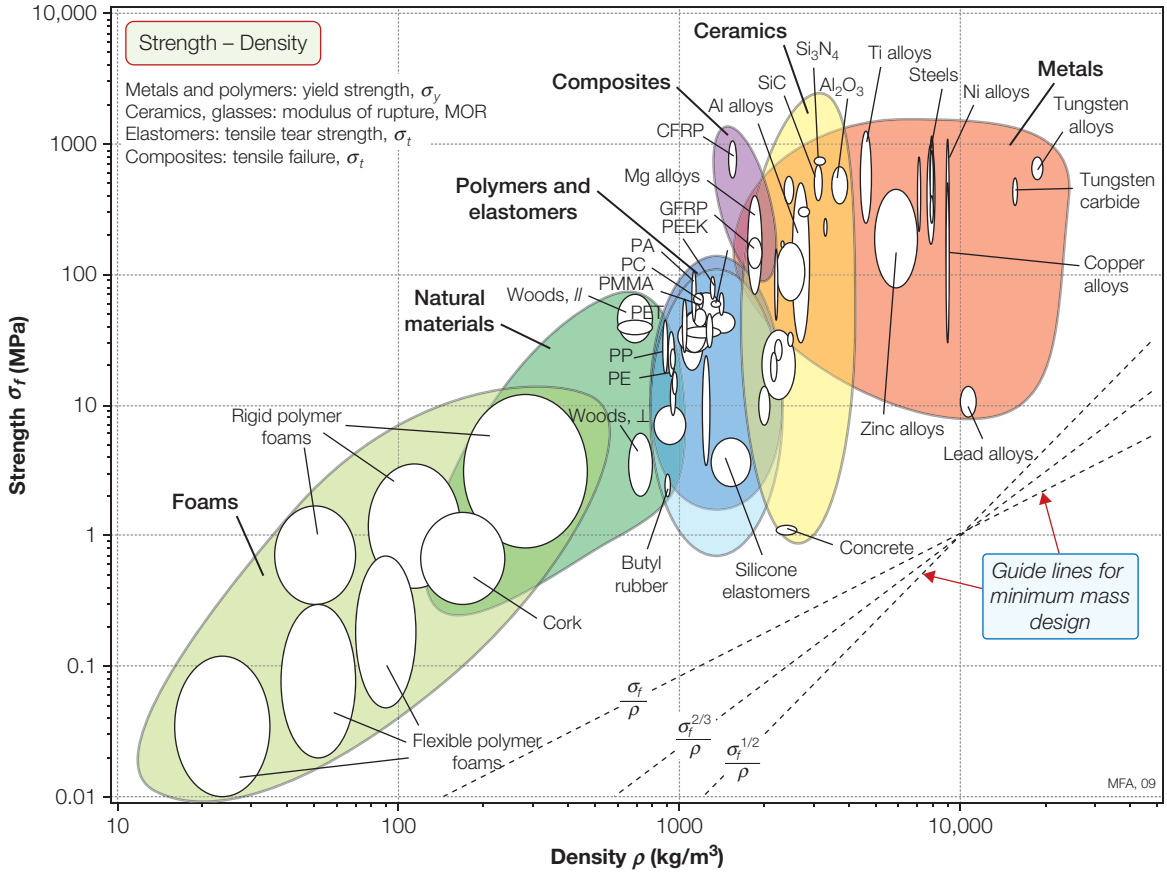


FIGURE 4.4

Strength σ_f plotted against density ρ (yield strength for metals and polymers, compressive strength for ceramics, tear strength for elastomers, and tensile strength for composites). The guide lines of constants σ_f/ρ, $\sigma_f^{2/3}/\rho$, and $\sigma_f^{1/2}/\rho$ are used in minimum weight, yield-limited, design.

the symbol σ_f for all of these, despite the different failure mechanisms involved, to allow a first-order comparison.

The range of strength for engineering materials, like the range for the modulus, spans many decades: from less than 0.01 MPa (foams, used in packaging and energy-absorbing systems) to 10^4 MPa (the strength of diamond, exploited in the diamond-anvil press). The single most important concept in understanding this wide range is the *lattice resistance* or *Peierls stress*. It is the intrinsic resistance of the structure to plastic shear. Plastic shear in a crystal involves the motion of dislocations. Pure metals are soft because the nonlocalized metallic bond does little to hinder dislocation motion, whereas ceramics are hard because their

more localized covalent and ionic bonds (which must be broken and reformed when the structure is sheared) lock the dislocations in place. In noncrystalline solids we think instead of the energy associated with the unit step of the flow process as the relative slippage of two segments of a polymer chain, or the shear of a small molecular cluster in a glass network. The strength of noncrystalline solids has the same origin as that underlying the lattice resistance. Thus if the unit step involves breaking strong bonds (as in an inorganic glass), the materials will be strong. If it only involves the rupture of weak bonds (the Van der Waals bonds in polymers for example), it will be weak. Materials that fail by fracture do so because the lattice resistance or its amorphous equivalent is so large that atomic separation (fracture) happens first.

When the lattice resistance is low, the material can be strengthened by introducing obstacles to slip. In metals this is achieved by adding alloying elements, particles, grain boundaries, and other dislocations ("work hardening"); in polymers, by cross-linking or by orienting the chains so that strong covalent bonds, as well as weak Van der Waals bonds, must be broken when the material deforms. When, on the other hand, the lattice resistance is high, further hardening is superfluous—the problem becomes that of suppressing fracture.

An important use of the chart is in materials selection for lightweight strength-limited design. Guide lines are shown for materials selection in the minimum-weight design of ties, columns, beams, and plates, and for yield-limited design of moving components in which inertial forces are important. Their use is described in Chapters 5 and 6.

The modulus–strength chart

High tensile steel makes good springs. But so does rubber. How is it that two such different materials are both suited to the same task? This and other questions are answered by Figure 4.5, one of the most useful of all the charts. It shows Young's modulus, E, plotted against strength, σ_f. The qualifications for "strength" are the same as before. It means yield strength for metals and

High strength at low weight

Which material has the highest ratio of strength σ_f to density ρ? Use Figure 4.4 to find out.

Answer

The materials with the largest values of σ_f/ρ are those toward the top left of the figure. The ratio plots as a line of slope 1 on the chart. There is a guide line with this slope among the three at the lower right. The materials with the highest ratio are the ones furthest above such a line. Carbon-fiber–reinforced polymers (CFRPs) stands out as meeting this criterion.

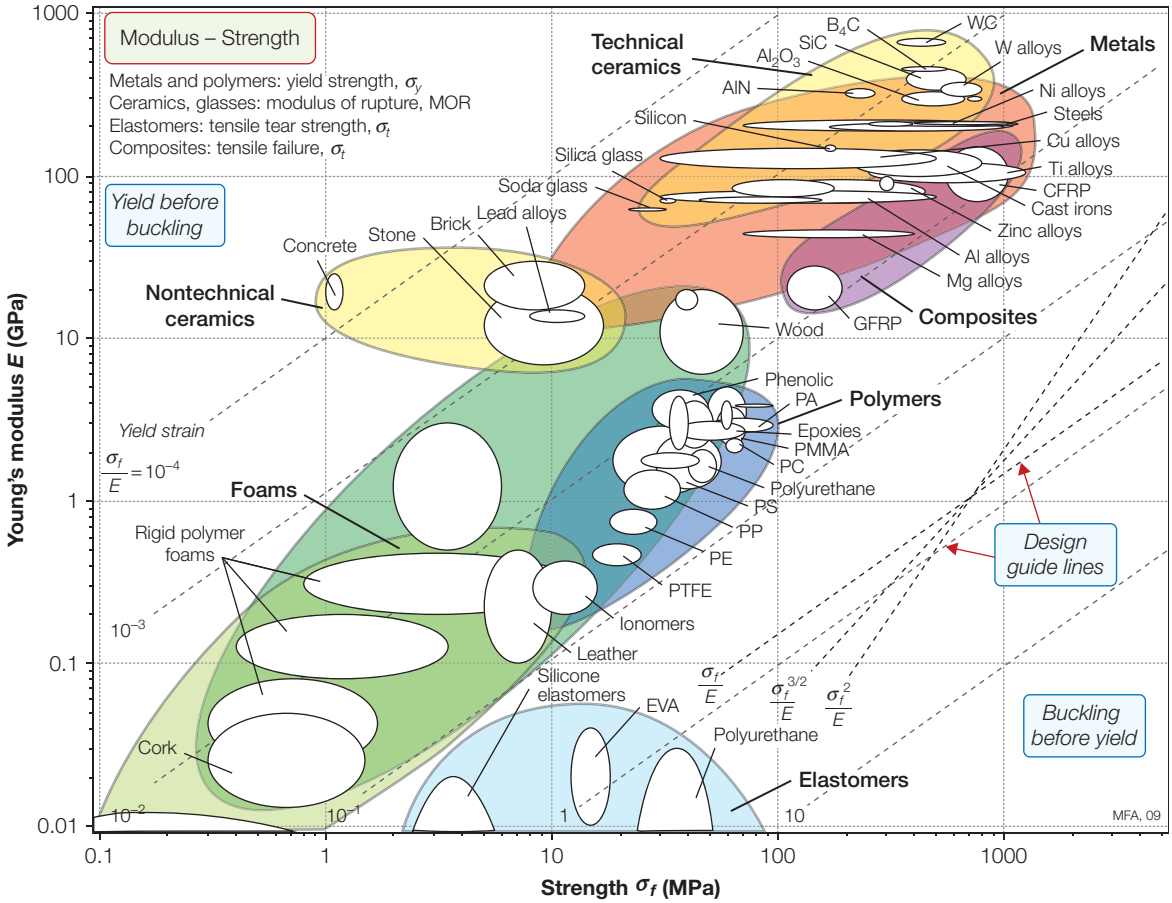


FIGURE 4.5

Young's modulus E plotted against strength σ_f. The design guide lines help with the selection of materials for springs, pivots, knife-edges, diaphragms, and hinges; their use is described in Chapters 5 and 6.

polymers, flexural strength (modulus of rupture) for ceramics, tear strength for elastomers, and tensile strength for composites and woods; the symbol σ_f is used for them all. Contours of *yield strain* or *fracture strain*, σ_f/E (meaning the strain at which the material ceases to be linearly elastic), appear as a family of straight parallel lines.

Examine these first. Engineering polymers have large yield strains of between 0.01 and 0.1; the values for metals are at least a factor of 10 smaller. Composites and woods lie on the 0.01 contour, as good as the best metals. Elastomers, because of their exceptionally low moduli, have values of σ_f/E larger than any other class of material: typically 1 to 10.

The distance over which inter atomic forces act is small; a bond is broken if it is stretched to more than about 10% of its original length. So the force F^* needed to break a bond is roughly

$$F^* \approx \frac{Sr_o}{10} \tag{4.3}$$

where S, as before, is the bond stiffness. Then the failure strain of a solid should be roughly

$$\frac{\sigma_f}{E} \approx \frac{F^*}{r_o^2} / \left(\frac{S}{r_o}\right) = \frac{1}{10} \tag{4.4}$$

The chart shows that, for some polymers, the failure strain approaches this value. For most solids it is less, for two reasons.

First, nonlocalized bonds (those in which the cohesive energy derives from the interaction of one atom with large number of others, not just with its nearest neighbors) are not broken when the structure is sheared. The metallic bond, and the ionic bond for certain directions of shear, act in this way. Very pure metals, for example, yield at stresses as low as $E/10{,}000$, and strengthening mechanisms are needed to make them useful in engineering. The covalent bond *is* localized, and covalent solids, for this reason, have yield strengths that, at low temperatures, are as high as $E/10$. It is difficult to measure them (though it can sometimes be done by indentation) because of the second reason for weakness: They generally contain defects—concentrators of stress—from which fractures can propagate at stresses well below the "ideal" $E/10$. Elastomers are anomalous (they have strengths of about E) because the modulus does not derive from bond stretching, but from the change in entropy of the tangled molecular chains when the material is deformed.

It has not yet explained how to choose good materials to make springs. This involves the design guide lines shown on the chart, and it will be examined closer in Section 6.7.

Strong solids

Use the strength-density chart of Figure 4.4 to identify three material classes with members having strengths exceeding 1,000 MPa.

Answer

The material classes of steels, titanium alloys, and carbon-fiber composites (CFRPs) all have members with strengths exceeding 1,000 MPa.

The specific stiffness–specific strength chart

Many designs, particularly those for things that move, call for stiffness and strength at minimum weight. To help with this, the data of the previous chart are replotted in Figure 4.6 after dividing, for each material, by the density; it shows E/ρ plotted against σ_f/ρ. These are measures of "mechanical efficiency," meaning the use of the least mass of material to do the most structural work.

Composites, particularly CFRP, lie at the upper right. They emerge as the material class with the most attractive specific properties, one of the reasons for their increasing use in aerospace. Ceramics have exceptionally high stiffness per unit weight, and their strength per unit weight is as good as

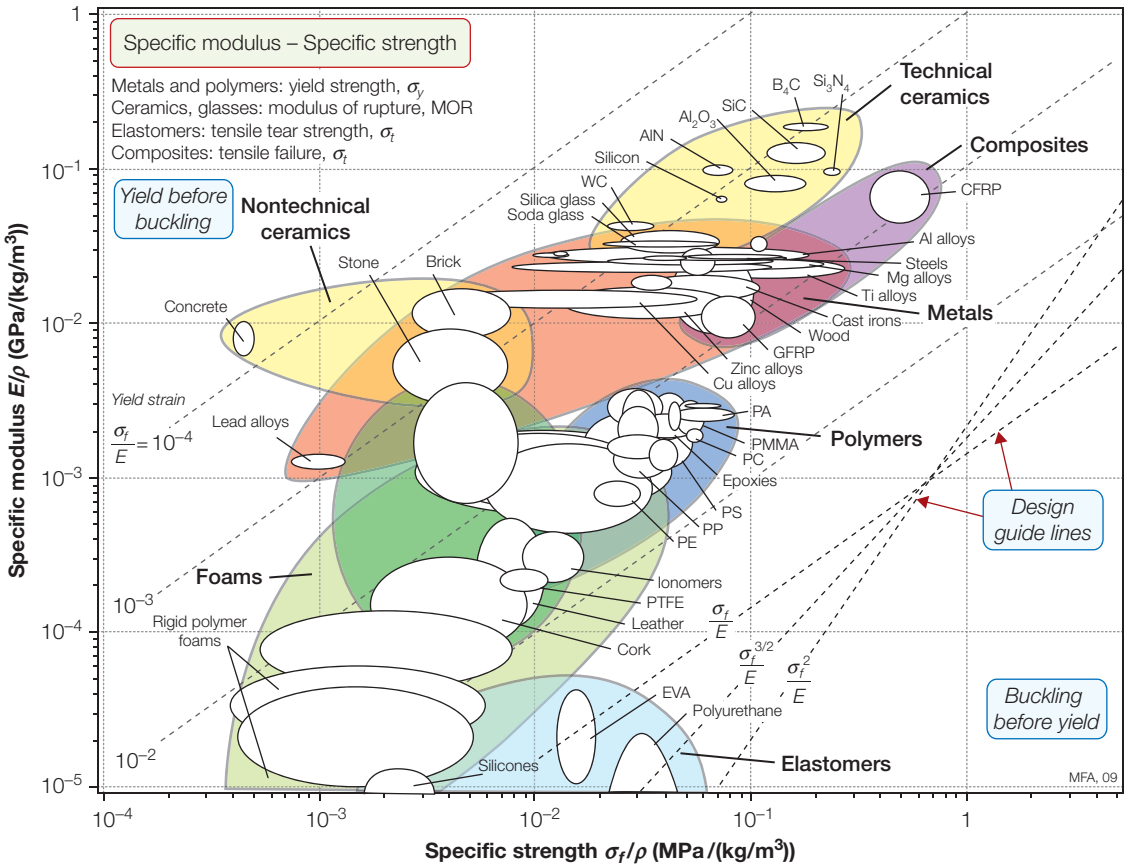


FIGURE 4.6

Specific modulus E/ρ plotted against specific strength σ_f/ρ. The design guide lines help with the selection of materials for lightweight springs and energy-storage systems.

High strength at low weight

High-quality mountain bikes are made of materials with particularly high values of the ratio σ_f/ρ, making them strong and light. Which metal class has the highest value of this ratio?

Answer

Figure 4.6 shows that titanium alloys have the highest value.

that of metals, but their brittleness excludes them from much structural use. Metals are penalized because of their relatively high densities. Polymers, because their densities are low, do better on this chart than on the last one.

The chart shown earlier in Figure 4.6 has application in selecting materials for light springs and energy storage devices. We will examine this in Section 6.7.

The fracture toughness–modulus chart

Increasing the strength of a material is useful only as long as the material remains plastic and does not become brittle; if it does, it is vulnerable to failure by fast fracture initiated from any tiny crack or defect it may contain. The resistance to the propagation of a crack is measured by the *fracture toughness*, K_{1c}, the units of which are MPa.m$^{1/2}$. It is plotted against modulus E in Figure 4.7. Values range from less than 0.01 to over 100 MPa.m$^{1/2}$. At the lower end of this range are brittle materials, which, when loaded, remain elastic until they fracture. For these, linear-elastic fracture mechanics works well, and the fracture toughness itself is a well-defined property.

At the upper end lie the super-tough materials, all of which show substantial plasticity before they break. For these the values of K_{1c} are approximate, derived from critical J-integral (J_c) and critical crack-opening displacement (δ_c) measurements, by writing $K_{1c} = (EJ_c)^{1/2}$, for instance. They are helpful in providing a ranking of materials. The figure shows one reason for the dominance of metals in engineering; they almost all have values of K_{1c} above 18 MPa.m$^{1/2}$, a value often quoted as a minimum for conventional design.

As a general rule, the fracture toughness of polymers is about the same as that of ceramics and glasses. Yet polymers are widely used in engineering structures; ceramics, because they are "brittle," are treated with much more caution. Figure 4.7 helps resolve this apparent contradiction. Consider first the question of the *necessary condition for fracture*. It is that sufficient external work be done, or elastic energy released, to supply the surface energy, γ per unit area, of the two new surfaces that are created. We write this as

$$G \geq 2\gamma \qquad (4.5)$$

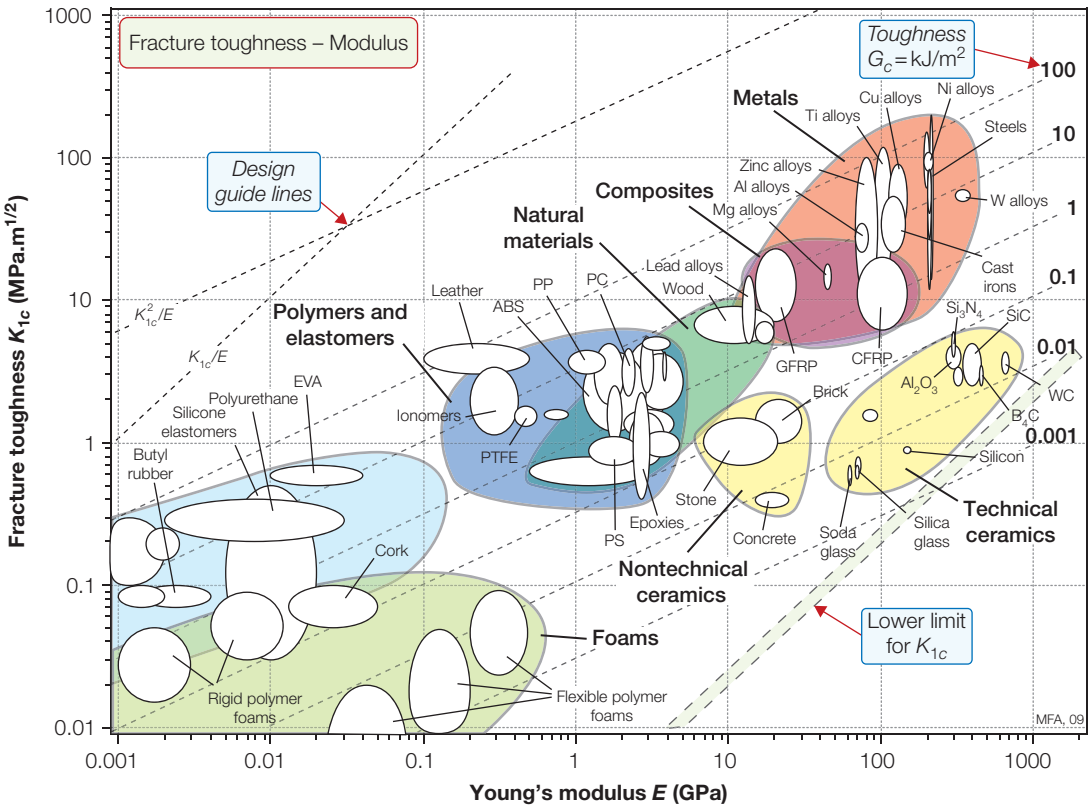


FIGURE 4.7

Fracture toughness K_{1c} plotted against Young's modulus E. The family of lines is of constant K_{1c}^2/E (approximately G_{1c}, the fracture energy or toughness). These, and the guide line of constant K_{1c}/E, help in design against fracture. The shaded band shows the lower limit for K_{1c}.

where G is the energy release rate. Using the standard relation $K = (EG)^{1/2}$ between G and stress intensity K, we find

$$K \geq (2E\gamma)^{1/2} \tag{4.6}$$

Now the surface energies, γ, of solid materials scale as their moduli; to an adequate approximation $\gamma \approx Er_o/20$, where r_o is the atom size, giving

$$K \geq E\left(\frac{r_o}{20}\right)^{1/2} \tag{4.7}$$

We identify the right-hand side of this equation with a lower-limiting value of K_{1c}, when, taking r_o as 2×10^{-10} m,

$$\frac{(K_{1c})_{\min}}{E} = \left(\frac{r_o}{20}\right)^{1/2} \approx 3 \times 10^{-6}\ \text{m}^{1/2} \tag{4.8}$$

Comparing materials by toughness

The fracture toughness K_{1c} of polypropylene (PP) is about 4 MPa.m$^{1/2}$. That of aluminum alloys is about 10 times greater. But in deflection-limited design it is toughness G_c that is the more important property. Use Figure 4.7 to compare the two materials by toughness.

Answer

Aluminum and PP have almost exactly the same values of G_c: approximately 10 kJ/m^2.

This criterion is plotted on the chart as a shaded, diagonal band near the lower right corner. It defines a *lower limit* for K_{1c}. The fracture toughness cannot be less than this unless some other source of energy such as a chemical reaction, or the release of elastic energy stored in the special dislocation structures caused by fatigue loading, is available, when it is given a new symbol such as $(K_1)_{scc}$ meaning "the critical value of K_1 for stress-corrosion cracking" or $(\Delta K_1)_{threshold}$ meaning "the minimum range of K_1 for fatigue-crack propagation." We note that the most brittle ceramics lie close to the threshold. When they fracture, the energy absorbed is only slightly more than the surface energy. When metals, polymers, and composites fracture, the energy absorbed is vastly greater, usually because of plasticity associated with crack propagation.

Plotted on Figure 4.7 are contours of *toughness*, G_c, a measure of the apparent fracture surface energy ($G_c \approx K_{1c}^2/E$). The true surface energies, γ, of solids lie in the range 10^{-4} to 10^{-3} kJ/m^2. The diagram shows that the values of the toughness start at 10^{-3} kJ/m^2 and range through almost five decades to over 100 kJ/m^2. On this scale, ceramics (10^{-3}–10^{-1} kJ/m^2) are much lower than polymers (10^{-1}–10 kJ/m^2); this is part of the reason polymers are more widely used in engineering than ceramics. This point is developed further in Section 6.10.

The fracture toughness–strength chart

The stress concentration at the tip of a crack generates a *process zone*: a plastic zone in ductile solids, a zone of micro-cracking in ceramics, and a zone of delamination, debonding, and fiber pull-out in composites. Within the process zone, work is done against plastic and frictional forces; it is this that accounts for the difference between the measured fracture energy, G_c, and the true surface energy 2γ. The amount of energy dissipated must scale roughly with the strength of the material within the process zone and with its size, d_γ. This size is found by equating the stress field of the crack ($\sigma = K/\sqrt{2\pi r}$) at $r = d_\gamma/2$ to the strength of the material, σ_f, giving

$$d_\gamma = \frac{K_{1c}^2}{\pi \sigma_f^2} \tag{4.9}$$

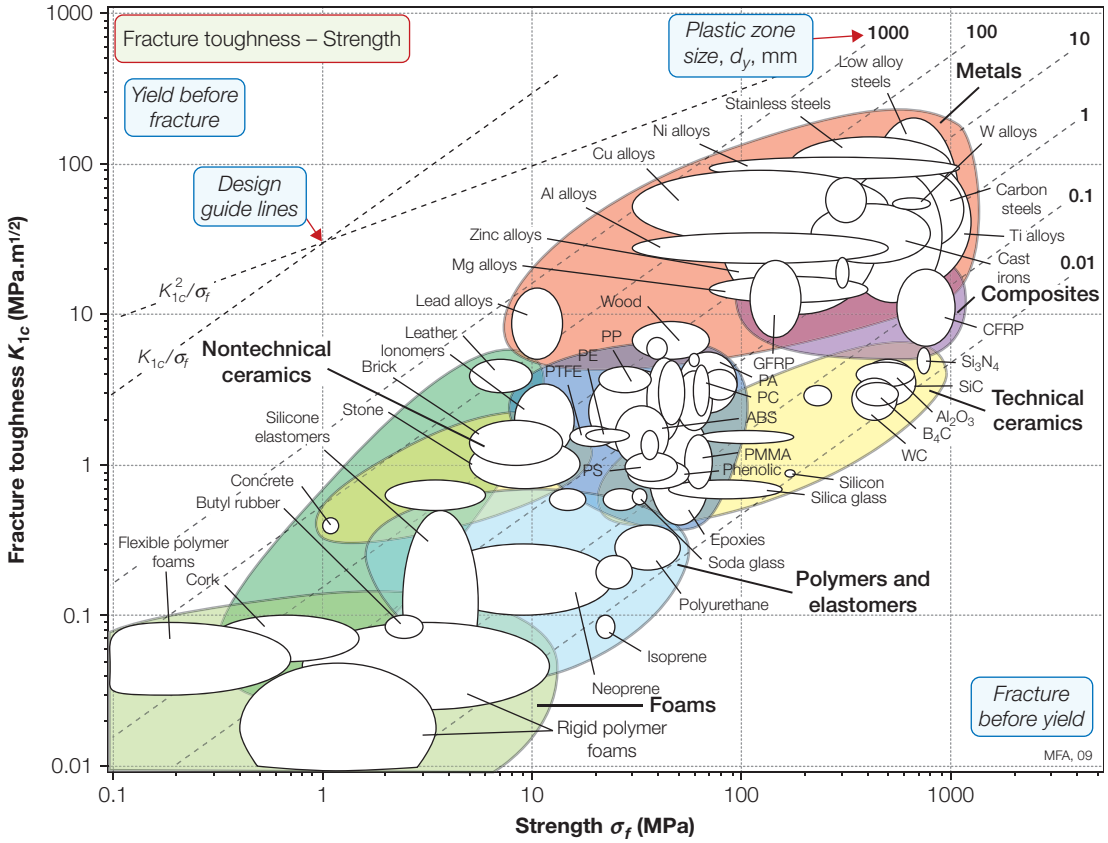


FIGURE 4.8

Fracture toughness K_{1c} plotted against strength σ_f. The contours show the value of $K_{1c}^2/\pi\sigma_f^2$—roughly the diameter d_y of the process zone at a crack tip. The design guide lines are used in selecting materials for damage-tolerant design.

Figure 4.8, fracture toughness plotted against strength, shows that the size of the zone, d_y (broken lines), varies from atomic dimensions for very brittle ceramics and glasses to almost 1 meter for the most ductile of metals. At a constant zone size, fracture toughness tends to increase with strength, as expected. It is this that causes the data plotted in Figure 4.8 to be clustered around the diagonal of the chart.

Materials toward the bottom right have high strength and low toughness; they *fracture before they yield*. Those toward the top left do the opposite: they *yield before they fracture*.

The diagram has application in selecting materials for the safe design of load-bearing structures. Examples are given in Sections 6.10 and 6.11.

Valid toughness testing

A valid fracture toughness test requires a sample with dimensions at least 10 times larger than the diameter of the process zone that forms at the crack tip. Use Figure 4.8 to estimate the sample size needed for a valid test on ABS.

Answer

The process zone size for ABS is approximately 1 mm. A valid test requires a sample with dimensions exceeding 10 mm.

The loss coefficient–modulus chart

Bells, traditionally, are made of bronze. They can be made of glass, and they could (if you could afford it) be made of silicon carbide. Metals, glasses, and ceramics all, under the right circumstances, have low intrinsic damping or "internal friction," an important material property when structures vibrate. Intrinsic damping is measured by the *loss coefficient*, η, which is plotted in Figure 4.9.

There are many mechanisms of intrinsic damping and hysteresis. Some (the "damping" mechanisms) are associated with a process that has a specific time constant; the energy loss is centered about a characteristic frequency. Others, the "hysteresis" mechanisms, are time-independent; they absorb energy at all frequencies. In metals a large part of the loss is hysteretic, caused by dislocation movement: It is high in soft metals like lead and pure aluminum. Heavily alloyed metals like bronze and high-carbon steels have low loss because the solute pins the dislocations; these are the materials for bells. Exceptionally high loss is found in the Mn-Cu alloys, because of a strain-induced martensite transformation, and in magnesium, perhaps because of reversible twinning. The elongated bubbles for metals span the large range made accessible by alloying and work hardening. Engineering ceramics have low damping because of the enormous lattice-resistance pins dislocations that are in place at room temperature.

Porous ceramics, on the other hand, are filled with cracks, the surfaces of which rub, dissipating energy when the material is loaded. The high damping of flake-graphite cast irons has a similar origin. In polymers, chain segments slide against each other when loaded; the relative motion dissipates energy. The ease with which they slide depends on the ratio of the ambient temperature, T, in this case room temperature, to the glass temperature, T_g, of the polymer. When $T/T_g < 1$, the secondary bonds are "frozen," the modulus is high, and the damping is relatively low. When $T/T_g > 1$, the secondary bonds have melted, allowing easy chain slippage; the modulus is

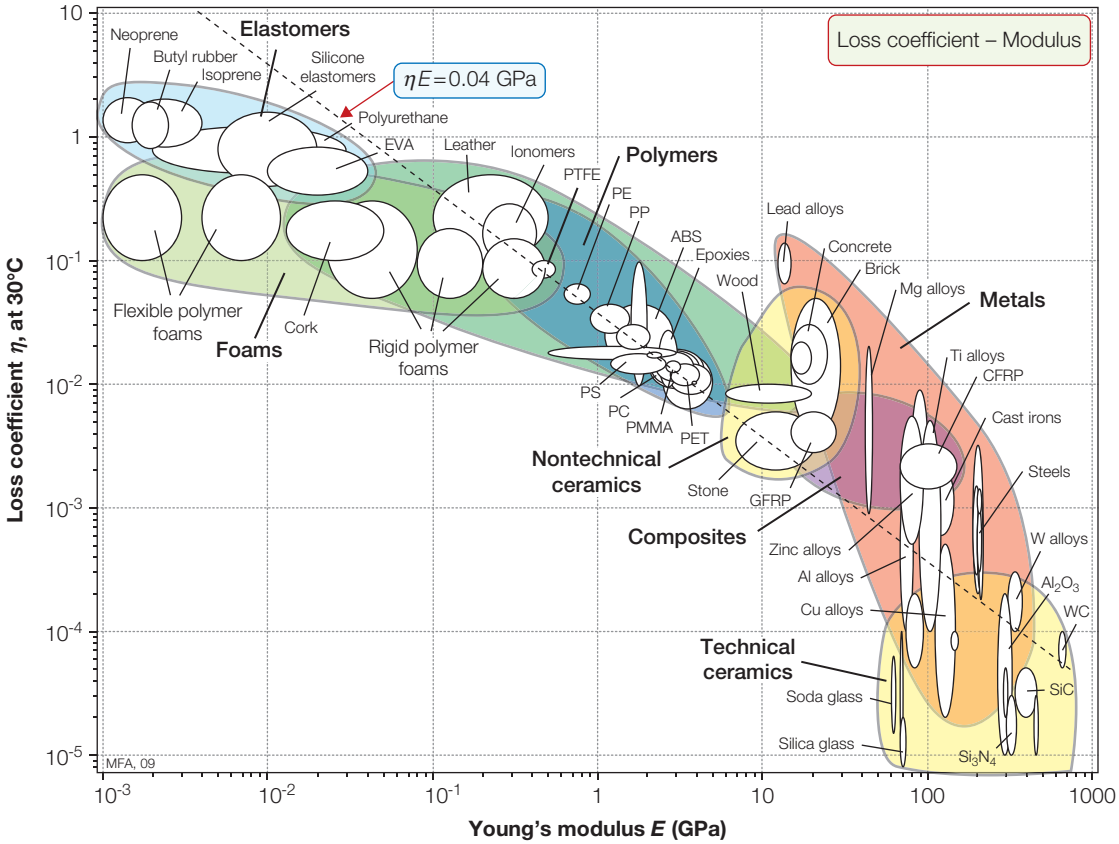


FIGURE 4.9
The loss coefficient η plotted against Young's modulus E. The guide line corresponds to the condition $\eta = CE$.

low, and the damping is high. This accounts for the obvious inverse dependence of η on E for polymers in Figure 4.9; indeed, to a first approximation,

$$\eta = \frac{4 \times 10^{-2}}{E} \tag{4.10}$$

(with E in GPa) for polymers, woods, and polymer-matrix composites.

Vibration damping

A metal is sought for mountings to damp vibration of a small machine tool. Use Figure 4.9 to find the metal with the greatest value of the damping coefficient η to use for the mountings.

Answer
Lead or lead alloys are the best choice.

Vibrational heating

A centrifuge rotating at f = 5,000 rpm is attached to a PTFE (Teflon) mounting. Poor balancing causes the centrifuge to vibrate, loading the PTFE up to a peak stress of σ_{max} = 8 MPa on each cycle. If a centrifuge run lasts 10 minutes and no heat is lost from the PTFE, by how much will its temperature rise? Take the volume-specific heat for PTFE (which can be read from Figure 4.11) to be $\rho C_p = 2 \times 10^6$ J/m^3.K, and retrieve the other material properties you need from Figure 4.9.

Answer

Figure 4.9 shows that the modulus of PTFE is E = 0.4 GPa and its loss coefficient is η = 0.08. The peak elastic energy stored in the PTFE in any one cycle is

$$U_{max} = \frac{\sigma_{max}^2}{2E} = 10^4 \text{ J/m}^3$$

of which $\Delta U = 2\pi\eta U_{max}$ (Equation (3.10)) is lost on each cycle. Thus the energy dumped into the PTFE over 5 minutes is

$$U_{10mins} = 2\pi\eta U_{max}(5f) = 2.52 \times 10^8 \text{ J/m}^3$$

Dividing this by the volume-specific heat of PTFE reveals a temperature rise of 126 °C. It will be necessary to ensure that heat can be conducted away from the mount to prevent it overheating.

The thermal conductivity–electrical resistivity chart

The material property governing the flow of heat through a material at steady state is the *thermal conductivity*, λ (units: W/m.K). (See Figure 4.10.) The valence electrons in metals are "free," moving like a gas within the lattice of the metal. Each electron carries a kinetic energy, $\frac{3}{2}kT$, where k is Boltzmann's constant. It is the transmission of this energy, via collisions, that conducts heat in metals. The thermal conductivity is described by

$$\lambda = \frac{1}{3} C_e \bar{c} \ell \qquad (4.11)$$

where C_e is the electron specific heat per unit volume, $\bar{c}$ is the electron velocity (2×10^5 m/s), and ℓ is the electron mean-free path, typically 10^{-7} m in pure metals. In heavily alloyed solid solutions (stainless steels, nickel-based superalloys, and titanium alloys) the foreign atoms scatter electrons, reducing the mean free path to atomic dimensions ($\approx 10^{-10}$ m), much reducing λ.

These same electrons, when in a potential gradient, drift through the lattice, giving electrical conduction. The electrical conductivity, κ, is measured here

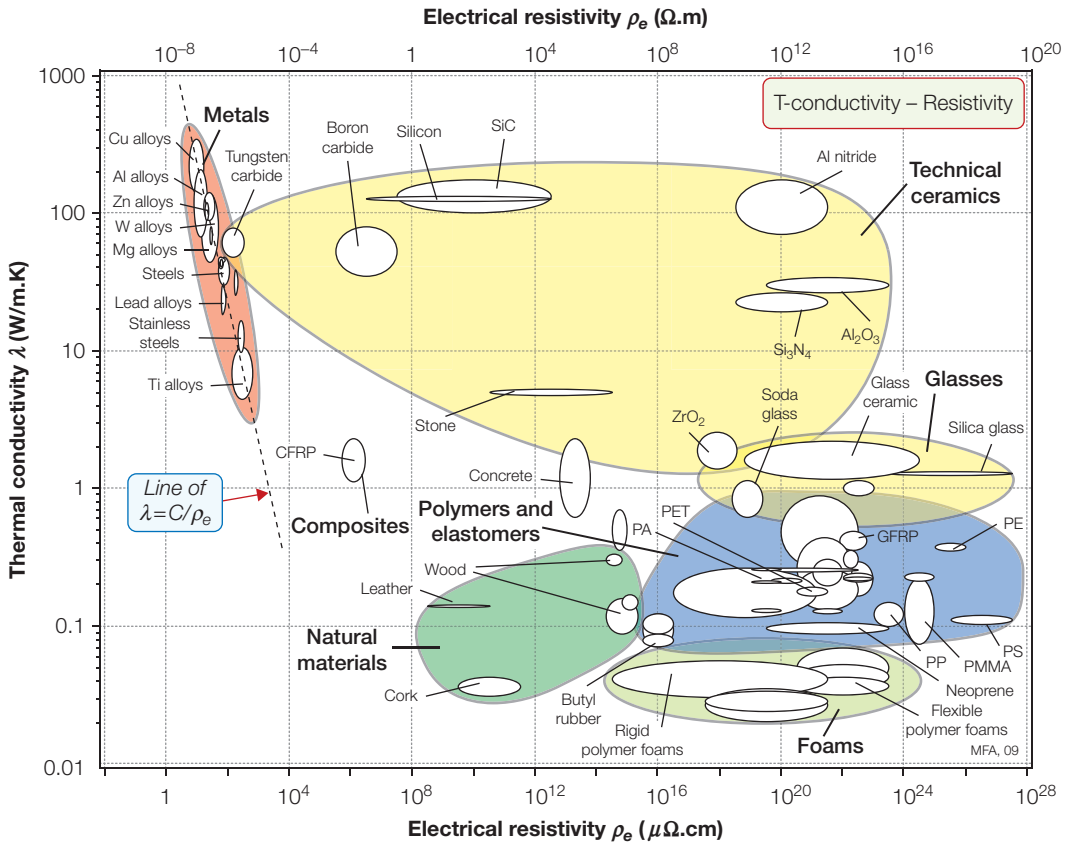


FIGURE 4.10
Thermal conductivity λ plotted against electrical resistivity ρ_e. For metals the two are related.

by its reciprocal, the *resistivity* ρ_e (SI units: Ω.m, units of convenience $\mu\Omega$.cm). The range is enormous: a factor of 10^{28}, far larger than that of any other property. As with heat, the conduction of electricity in metals is proportional to the density of carriers, the electrons, and their mean-free path, leading to the Wiedemann-Franz relation

$$\lambda \propto \kappa = \frac{1}{\rho_e} \tag{4.12}$$

The quantities λ and ρ_e are the axes of Figure 4.10. Data for metals appear at the top left. The broken line shows that the Wiedemann-Franz relation is well obeyed.

But what about the rest of the chart? Electrons do not contribute to thermal conduction in ceramics and polymers. Heat is carried instead by

phonons—lattice vibrations of short wavelength. They are scattered by each other and by impurities, lattice defects, and surfaces; it is these that determine the phonon mean-free path, ℓ. The conductivity is still given by equation (4.11), which we write as

$$\lambda = \frac{1}{3}\rho C_p \bar{c} \ell \tag{4.13}$$

but now $\bar{c}$ is the elastic wave speed (around 10^3 m/s—see Figure 4.3), ρ is the density, and C_p is the *specific heat per unit mass* (units: J/kg.K). If the crystal is particularly perfect and the temperature is well below the Debye temperature, as in diamond at room temperature, the phonon conductivity is high: It is for this reason that single crystal silicon carbide and aluminum nitride have thermal conductivities almost as high as copper.

The low conductivity of glass is caused by its irregular amorphous structure; the characteristic length of the molecular linkages (about 10^{-9} m) determines the mean free path. Polymers have low conductivities because the elastic wave speed $\bar{c}$ is low (see Figure 4.3), and the mean free path in the disordered structure is small. Highly porous materials like firebrick, cork, and foams show the lowest thermal conductivities, limited by the thermal conductivity of the gas in their cells.

Graphite and many intermetallic compounds such as C and B_4C, like metals, have free electrons, but the number of carriers is smaller and the resistivity is higher than in metals. Defects such as vacancies and impurity atoms in ionic solids create positive ions that require balancing electrons. These can jump from ion to ion, conducting charge, but slowly because the carrier density is low. Covalent solids and most polymers have no mobile electrons and are insulators ($\rho_e > 10^{12}\,\mu\Omega$.cm)—they lie on the right side of Figure 4.10.

Under a sufficiently high potential gradient, anything will conduct. The gradient tears electrons free from even the most possessive atoms, accelerating them into collision with nearby atoms, knocking out more electrons and creating a cascade. The critical gradient is the *breakdown potential*, V_b (units: MV/m), defined in Chapter 3.

Conducting heat but not electricity

Which materials are both good thermal conductors and good electrical insulators (an unusual combination)? Use Figure 4.10 to find out.

Answer

The chart identifies aluminum nitride, alumina, and silicon nitride as having these properties. They are the ones at the top right.

The thermal conductivity–thermal diffusivity chart

Thermal conductivity, as we have said, governs the flow of heat through a material at steady state. The property governing transient heat flow is the *thermal diffusivity*, a (units: m^2/s). The two are related by

$$a = \frac{\lambda}{\rho C_p} \tag{4.14}$$

where ρ in kg/m^3 is the density. The quantity ρC_p is the *volumetric specific heat* (units: $J/m^3.K$). Figure 4.11 relates thermal conductivity, diffusivity, and volumetric specific heat, at room temperature.

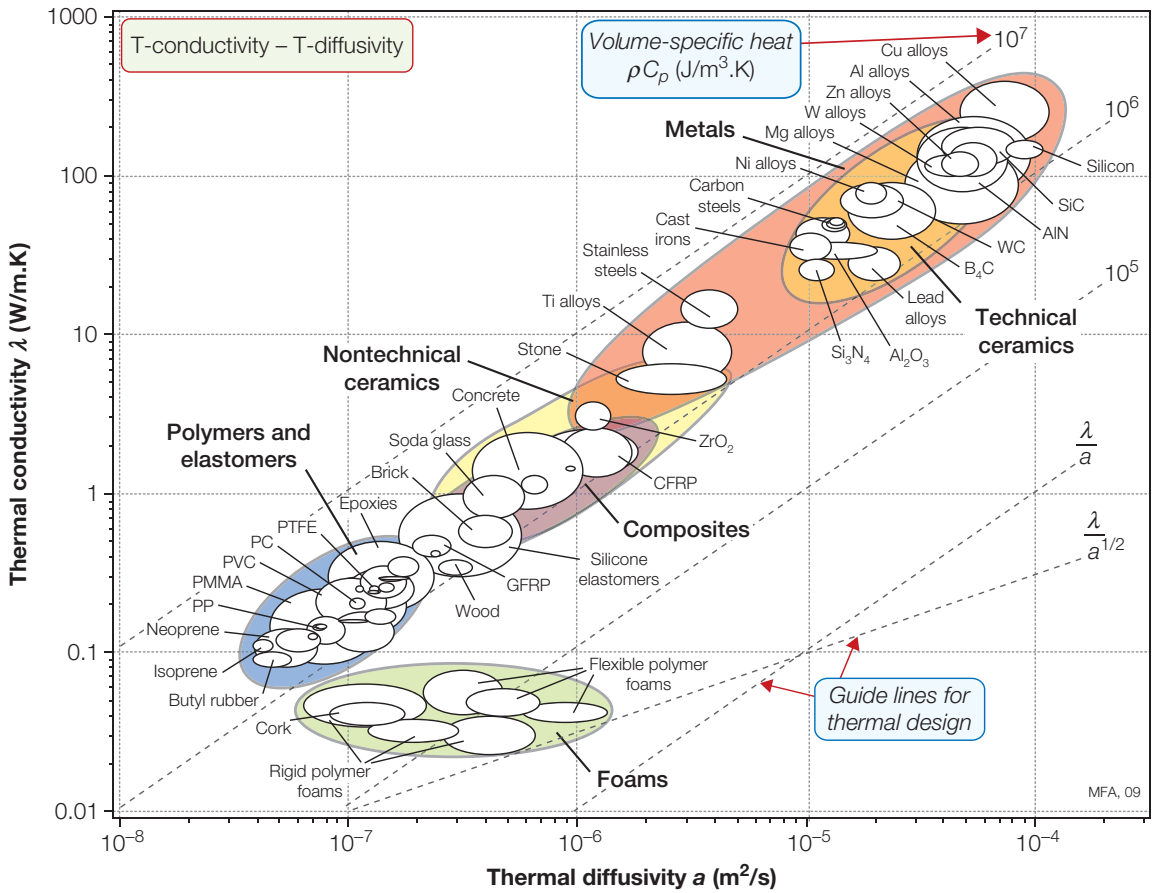


FIGURE 4.11

Thermal conductivity λ plotted against thermal diffusivity a. The contours show the volume-specific heat ρC_v. All three properties vary with temperature; the data here are for room temperature.

Thermal buffers

A good way to protect equipment from sudden temperature change is to encase it in a material with a very low thermal diffusivity, because an external temperature change then takes a long time to reach the inside. Use Figure 4.11 to identify materials that could be good for this.

Answer

The chart identifies isoprene, neoprene, and butyl rubber as potential candidates.

The data span almost five decades in λ and a. Solid materials are strung out along the line[3]

$$\rho C_p \approx 3 \times 10^6 \text{ J/m}^3\text{.K} \tag{4.15}$$

As a general rule, then,

$$\lambda = 3 \times 10^6 a \tag{4.16}$$

(λ in W/m.K and a in m^2/s). Some materials deviate from this rule because they have lower than average volumetric specific heat. The largest deviations are shown by porous solids: foams, low-density firebrick, woods, and the like. Their low density means that they contain fewer atoms per unit volume and, averaged over the volume of the structure, ρC_p is low. The result is that, although foams have low thermal *conductivities* and are widely used for insulation, their thermal *diffusivities* are not necessarily low: They may not transmit much heat, but they reach a steady state quickly. This is important in design, a point illustrated by the case study in Section 6.13.

The thermal expansion–thermal conductivity chart

Almost all solids expand on heating (Figure 4.12). The bond between a pair of atoms behaves like a linear elastic spring when the relative displacement of the atoms is small, but when it is large the spring is nonlinear. Most bonds become stiffer when the atoms are pushed together and less stiff when they are pulled apart. Such bonds are anharmonic. The thermal vibrations of atoms, even at room temperature, involve large displacements; as

[3] This can be understood by noting that a solid containing N atoms has $3N$ vibrational modes. Each (in the classical approximation) absorbs thermal energy kT at the absolute temperature T, and the vibrational specific heat is $C_p \approx C_v = 3Nk$ (J/K) where k is Boltzmann's constant (1.34×10^{-23} J/K). The volume per atom Ω for almost all solids lies within a factor of two of 1.4×10^{-29} m^3; thus the volume of N atoms is (NC_p) m^3. The volume-specific heat is then (as the chart shows):

$$\rho C_v \cong 3Nk/N\Omega = \frac{3k}{\Omega} = 3 \times 10^6 \text{ J/m}^3\text{K}$$

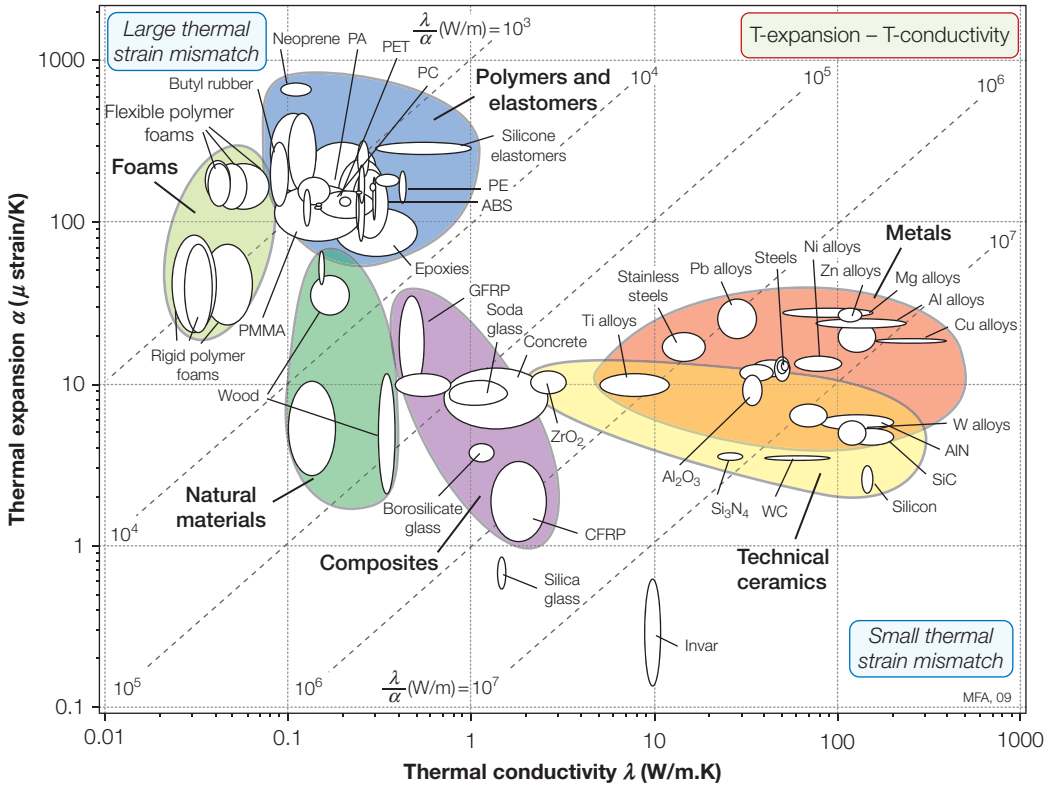


FIGURE 4.12

The linear expansion coefficient α plotted against the thermal conductivity λ. The contours show the thermal distortion parameter λ/α. An extra material, the nickel alloy Invar, has been added to the chart; it is noted for its exceptionally low expansion at and near room temperature, useful in designing precision equipment that must not distort if the temperature changes.

the temperature is raised, the anharmonicity of the bond pushes the atoms apart, increasing their mean spacing. The effect is measured by the linear *expansion coefficient*

$$\alpha = \frac{1}{\ell}\frac{d\ell}{dT} \tag{4.17}$$

where ℓ is a linear dimension of the body.

The expansion coefficient is plotted against the thermal conductivity in Figure 4.12. It shows that polymers have large values of α, roughly 10 times greater than those of metals and almost 100 times greater than those of ceramics. This is because the Van der Waals bonds of the polymer are very anharmonic. Diamond, silicon, and silica glass (SiO_2) have covalent bonds that have low anharmonicity (that is, they are almost linear elastic even at